



Xilinx Solutions for Medical & Scientific



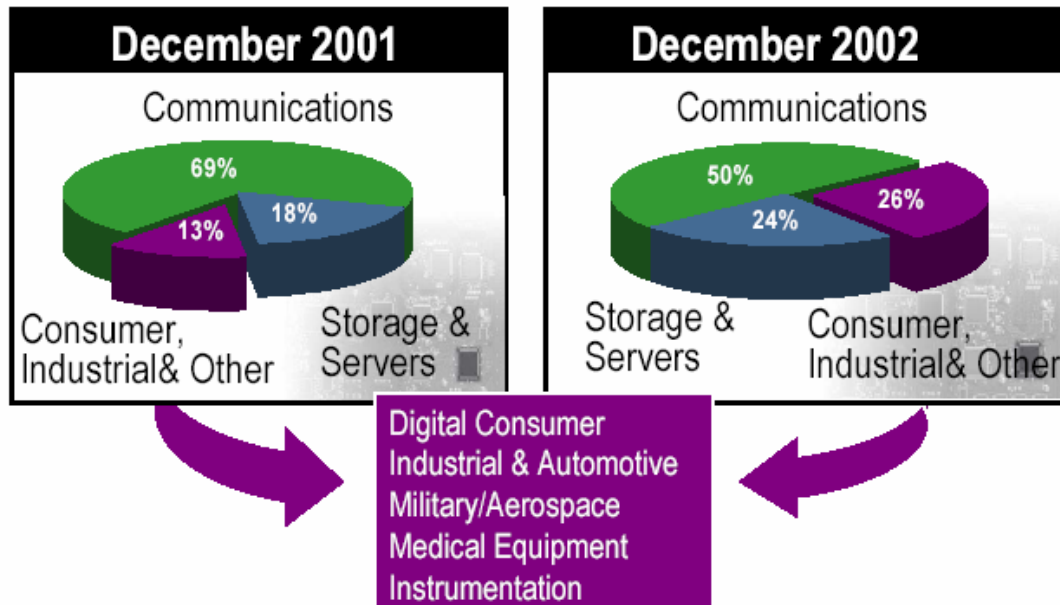
The Xilinx Value Proposition

- To enable worldwide customers to attain the fastest time-to-market and flexible product life cycle management by focusing on programmable logic solutions consisting of industry leading silicon, software and IP, and services resulting in a financially sound company.



Xilinx in Industrial, Medical & Scientific

- Industrial market – a key focus for Xilinx
 - Most stable portion of our business
- Broad base of customers worldwide



Medical Equipment and Instrumentation

- Imaging
 - Ultrasound, X-Ray, MRI, CT, nuclear
- Cardiac rhythm mgmt
 - ICDs, external defibrillators
- Home health
 - Breath alcohol testers, cholesterol level monitors, emergency ventilators
- Hospital equipment
 - Patient monitors
 - Dialysis machines
 - Infusion pumps
 - Endoscopy
 - Optometric
 - Powered beds
 - Physiotherapy equipment
 - Powered surgical instruments

Scientific and Lab Instrumentation

- Spectrometers
- Spectrophotometers
- Chromatography
- Sterilizers and autoclaves
- Centrifuges
- Incubators
- Laboratory balances

Issues & Requirements

- Image quality
 - Diagnosis often heavily based on captured images
 - Very high-resolution images need lots of processing power
 - Filtering for noise removal and enhancement
- Real-time digital processing
 - Doctors don't have to wait for film processing anymore
 - Can perform real-time image adjustments to aid investigations
- Keep it digital!
 - A-D and D-A conversion processes kept to a minimum to keep image quality high
 - Novel image processing techniques easier in digital domain

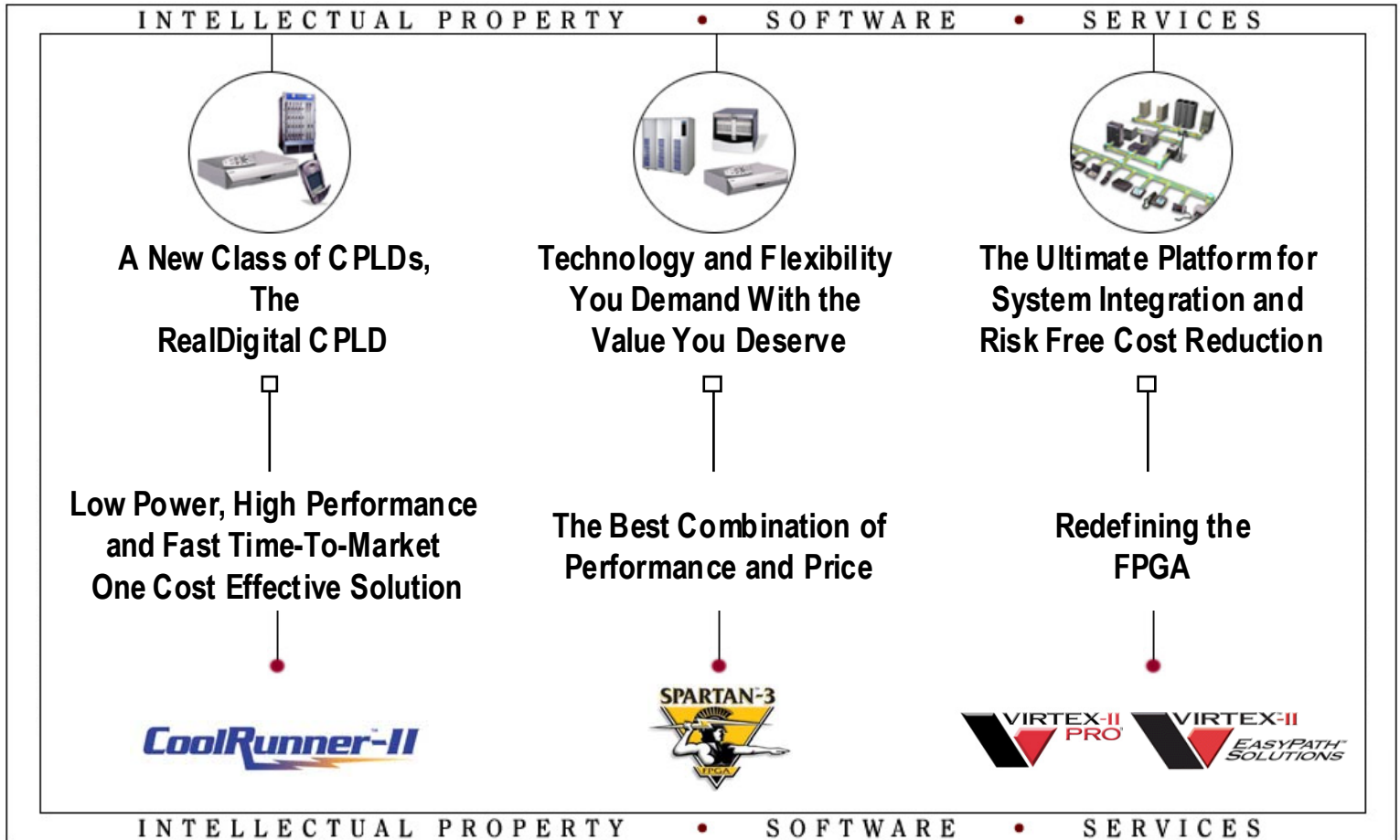
Experimenting with Tradeoffs

- It would be nice to have a fully flexible device to use for video processing designs
 - Allows changing of parameters like colour depth, bit accuracy (truncation)
 - Allows exploration of new compression techniques or acceleration of existing algorithms to improve throughput
 - Supports various frame rates and resolutions
 - Implements a wide range of new or existing filters for enhancement or noise reduction

Welcome to Xilinx FPGAs

- FPGAs are a key enabling technology for digital video processing
- Allow experimentation for prototypes leading to differentiation for production
- **And** still enable a higher level of system integration with support for
 - Video interfaces, LAN/WAN technologies, other DSP, simple glue, memory control and state machines, backplane protocols..... the list is only limited by the imagination

Product Solutions





Introducing Spartan-3



Xilinx' Latest Solution for the Medical
and Scientific Markets



The Spartan-3 Platform: A New Class of Spartan FPGAs

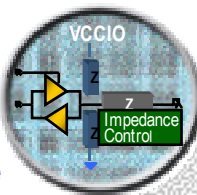
Advanced FPGA
Logic with Staggered
Pad Technology

90 nm

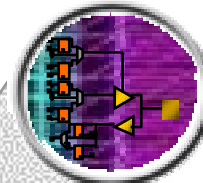
Embedded
XtremeDSP
Functionality

18 Bit
X 36 Bit
18 Bit

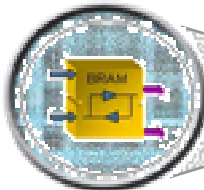
XCITE
Digitally
Controlled
Impedance



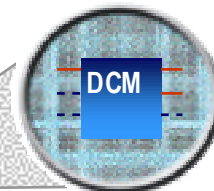
SelectIO™-Ultra
Technology



High Performance
Sync Dual-Port™
RAM



DCM™ Digital
Clock
Management



Alliance
CORE™

LogiCORE™

ISE5

XPERTS



Xilinx Global Services

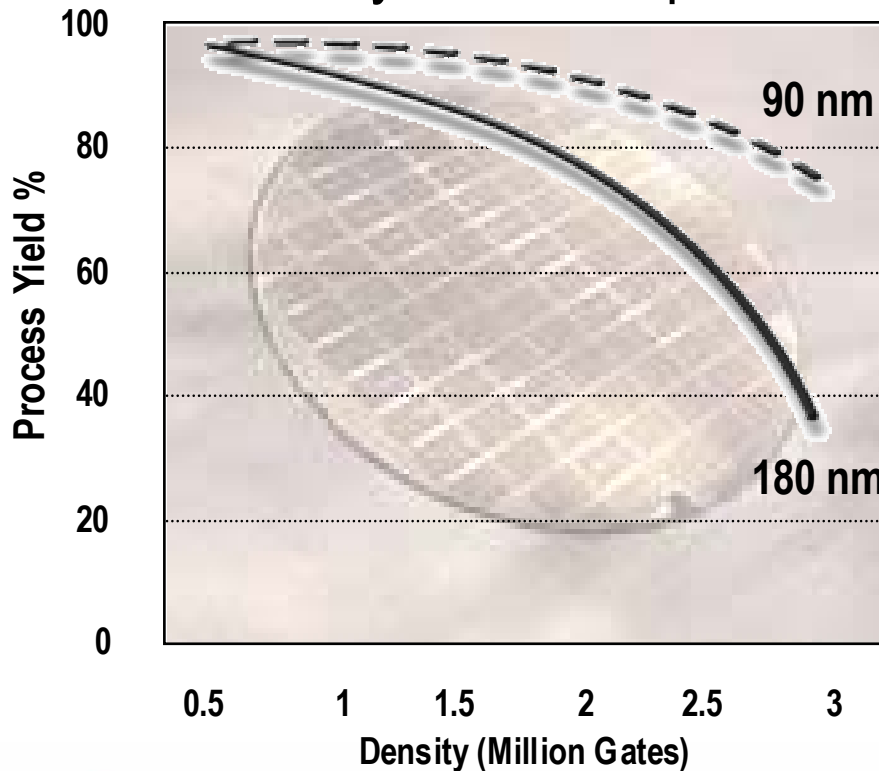
esp
EMERGING STANDARDS
& PROTOCOLS



Highest Density with Low Cost

90 nm Process Delivers Higher Yields at Higher Densities

Density Leadership



"The companies that get into 90nm production first will get a tremendous advantage in lower cost..... Rivals who are late in 90nm process technology will fall behind and may not be able to catch up."

Dan Hutcheson
VLSI Research, Inc.

- Faster performance
- Large density devices
- Lower cost devices

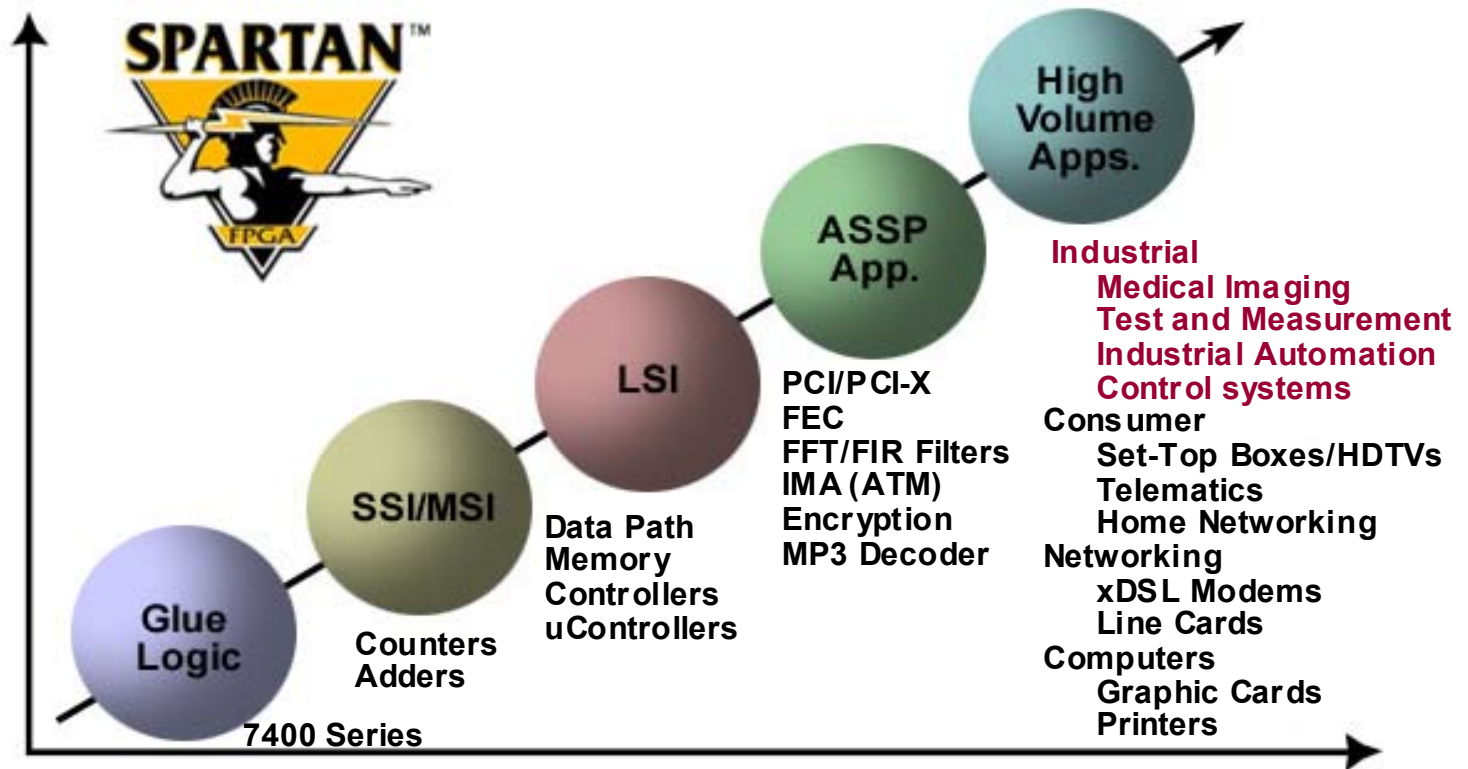
Spartan-3 Product Matrix

Device	XC3S50	XC3S200	XC3S400	XC3S1000	XC3S1500	XC3S2000	XC3S4000	XC3S5000
System Gates	50K	200K	400K	1000K	1500K	2000K	4000K	5000K
Logic Cells	1,728	4,320	8,064	17,280	29,952	46,080	62,208	74,880
Dedicated Multipliers	-	12	16	24	32	40	96	104
Block RAM Blocks	-	12	16	24	32	40	96	104
Block RAM Bits	-	216K	288K	432K	576K	720K	1,728K	1,872K
Distributed RAM Bits	12K	30K	56K	120K	208K	320K	432K	520K
DCMs	-	4	4	4	4	4	4	4
I/O Standards	24	24	24	24	24	24	24	24
Max Differential I/O	56	76	116	175	221	270	312	344
Max Single Ended I/O	124	173	264	391	487	565	712	784
TQ144 22x22mm	97							
PQ208 30.6x30.6mm	124	141	141					
FT256 17x17mm		173	173	173				
FG456 23x23mm			264	333	333			
FG676 27x27mm				391	487	489		
FG900 31x31mm						565	633	633

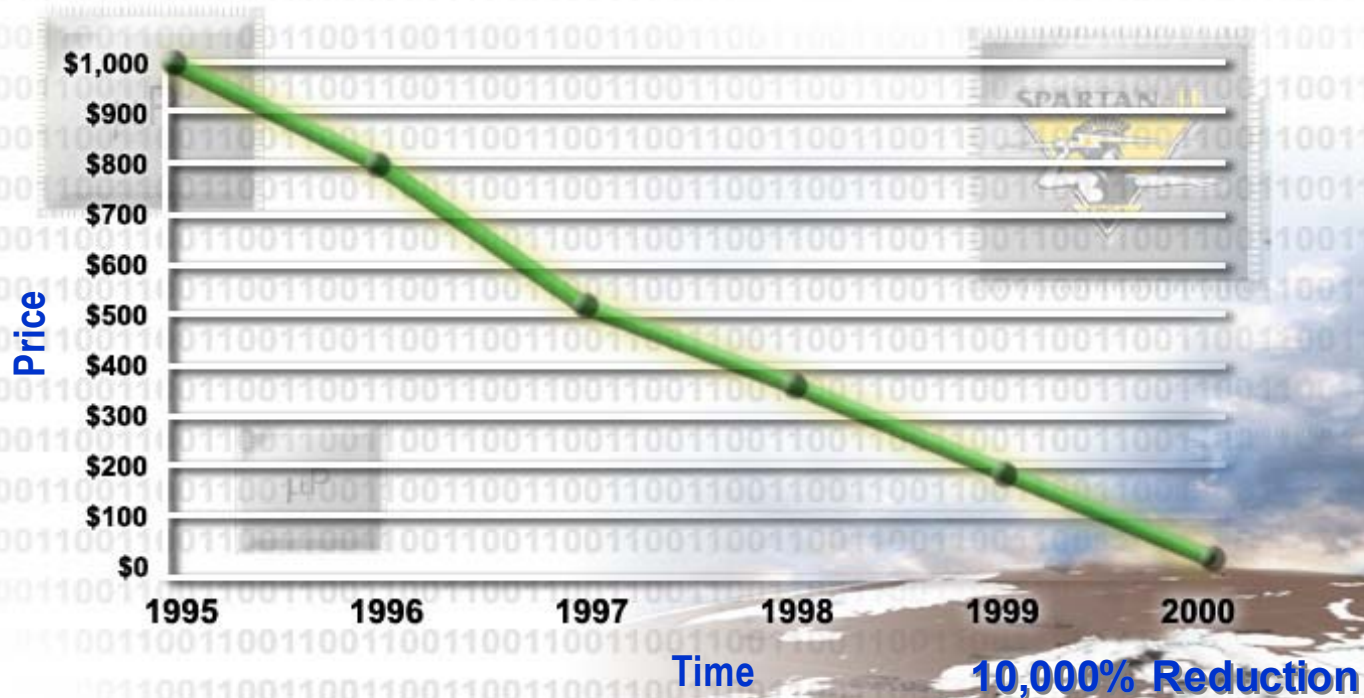
Spartan-3 Summary

Spartan-3 Feature	Value
FPGA Fabric and Routing, Up to 5,000,000 System Gates	Allows for implementation of system level function blocks, high on-chip connectivity and high-throughput
BlockRAM - 18K Blocks	Enables implementation of large packet buffers/FIFOs, line buffers
Distributed RAM	For implementing smaller FIFOs/Buffers, DSP coefficients
Shift Register Mode (SRL16E)	16-bit Shift Register ideal for capturing high speed or burst mode data and to store data in DSP and Encryption applications
Dedicated 18 x 18 Multiplier Blocks	High speed DSP processing; Use of multipliers in conjunction with fabric allows for ultra-fast, parallel DSP operations
Single-ended Signaling - GTL, GTL+, PCI; HSTL-I, II, III; SSTL3-I, II; SSTL2-I, II;	Connectivity to commonly used chip-to-chip, memory (SRAM, SDRAM) and chip-to-backplane signaling standards; eliminates the need for multiple level translators
Differential Signaling (up to 622 Mbps) - LVDS, BLVDS, Ultra LVDS, RSDS, LDT	Differential signaling at low cost - Bandwidth management (saving the number of pins), reduced power consumption, reduced EMI, high noise immunity
Digital Clock Management (DCM)	Eliminate on-chip & board-level clock delay, simultaneous multiply and divide, reduction of board-level clock speed and no. of board-level clocks, adjustable clock phase for ensuring coherency
Global routing resources	Distribution of clocks and other signals with very high fanout throughout the device
Programmable output drive	Improves signal integrity; achieving right tradeoff between Tco and ground bounce

PLD Application Trends



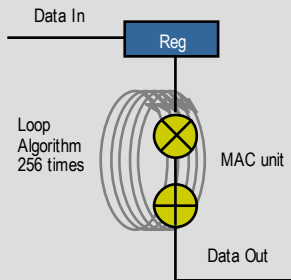
FPGA Price Reduction



Price of 100K FPGA Gates Over Time

Enabling High-Speed DSP

Conventional DSP/Software

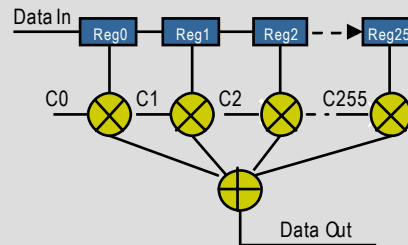


- Fixed inflexible architecture
 - Typically 1-4 MAC units
 - Fixed data width
- Serial processing limits data throughput
 - Time-shared MAC unit
 - High clock frequency creates difficult system-challenge

e.g. 256 Tap FIR Filter = 256 MAC operations per sample
256 MAC operations in 256 clock cycles in DSP!!

**Performance
Limitations**

FPGA Performance Advantage



- Flexible architecture
 - Distributed DSP resources (LUT, registers, multipliers, & memory)
- Parallel processing maximizes data throughput
 - Support any level of parallelism
 - Optimal performance/cost tradeoff

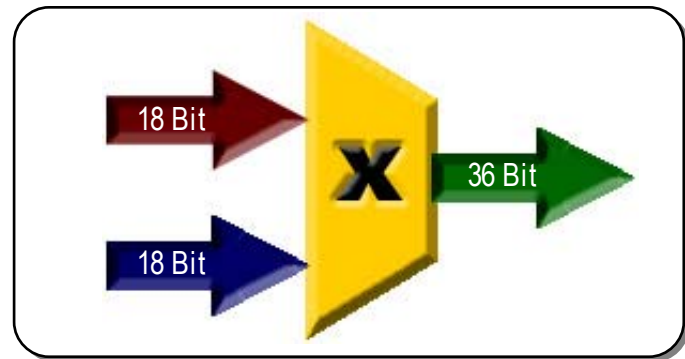
e.g. 256 Tap FIR Filter = 256 MAC operations per sample
All 256 MAC operations in 1 clock cycle in FPGA !!



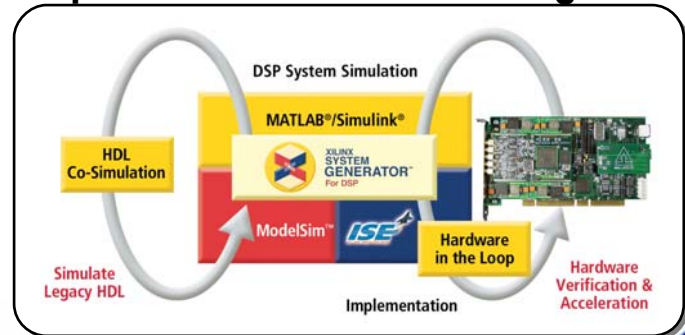
Unrivaled DSP Performance per Dollar

- Nearly 3 billion MACs/sec per dollar
- Cost effective solution for low cost DSP applications such as:
 - Digital communications, video/imaging, & industrial control
- Simple design flow
 - MathWorks (MATLAB/Simulink)
- Complete DSP solution
 - Silicon, software, IP, services, specialists & development systems

Embedded DSP Capability Up to 104 18bit Multipliers!



Simple and Familiar DSP Design Flow

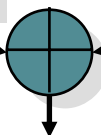


The Best of Both Worlds

- Off the shelf devices
- Faster time-to-market
- Rapid adoption of standards
- Real-time prototyping
- Parallel processing
- Support high data rates
- Optimal bit widths
- No real-time software coding

Flexibility of DSP Processors

Performance of Custom ICs

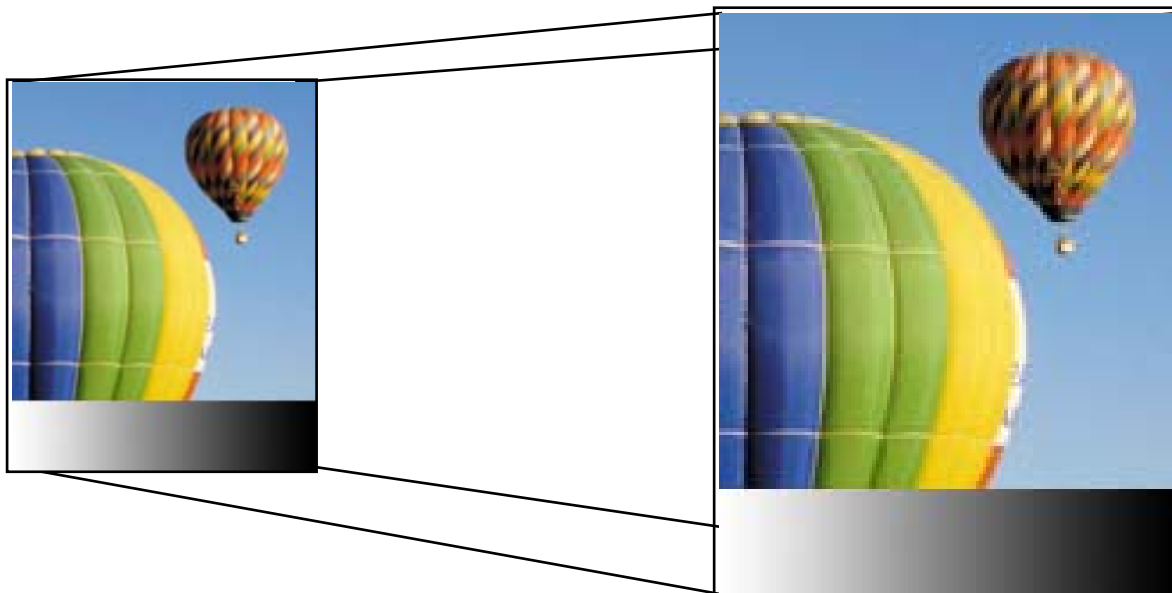


Xilinx DSP Solutions Offer the Best of Both Worlds at a Low Cost!

Basic Image Processing

Scaling

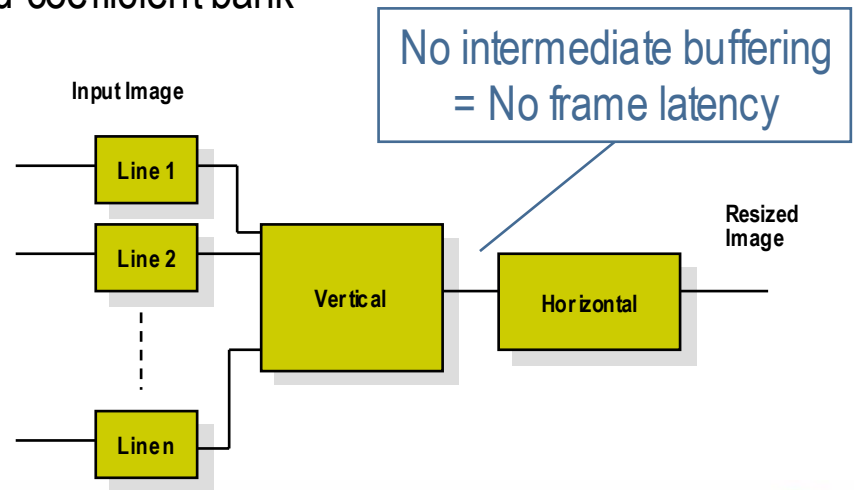
- Fractionally enlarges the incoming data stream as necessary to match the target display resolution
- Pixel processing on-the-fly during image input without frame buffer



Real-Time Image Resizing

With Low Memory Requirements 2 Dimensional Architecture Upscaling by 2, Downscaling by 4

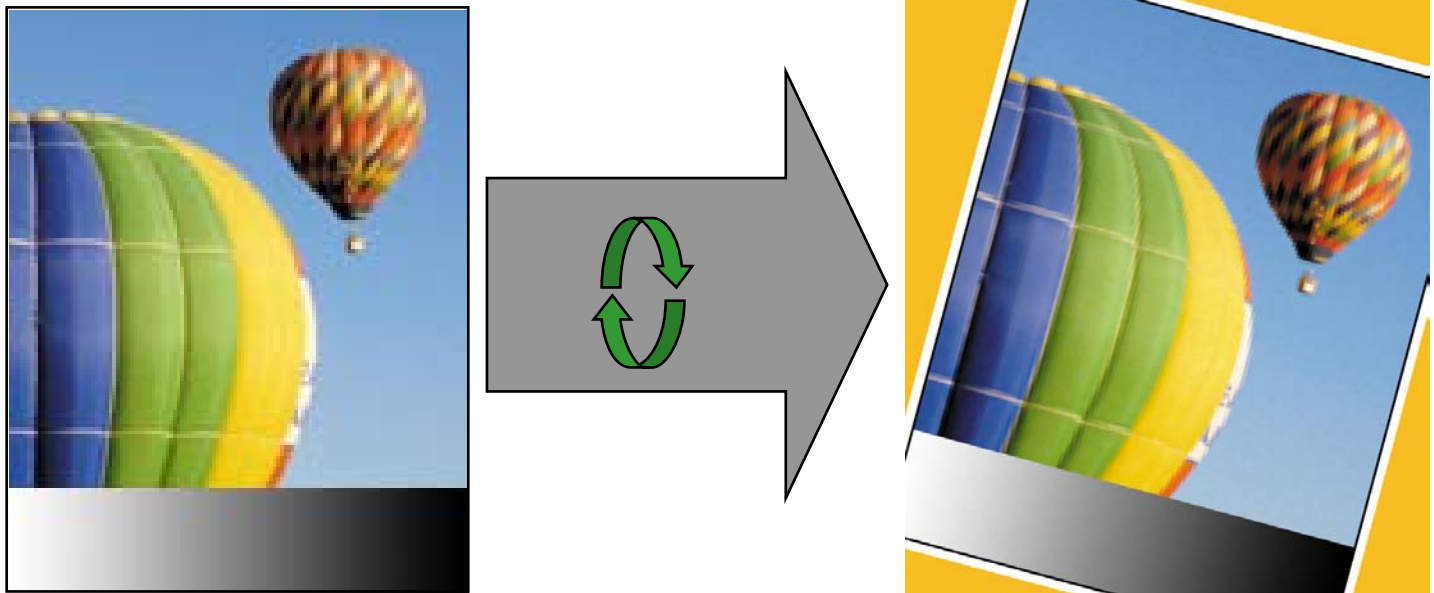
- Example: 512 x 512 x 8 60 f/s
 - Upscaling by 2, downscaling by 4
 - 16 pixel resolution
 - 8 Block RAMs for line buffers and coefficient bank
 - 4 vertical multipliers
 - 4 horizontal multipliers
 - Adder trees
 - Control



http://www.xilinx.com/products/logicore/dsp/rotation_resize.pdf

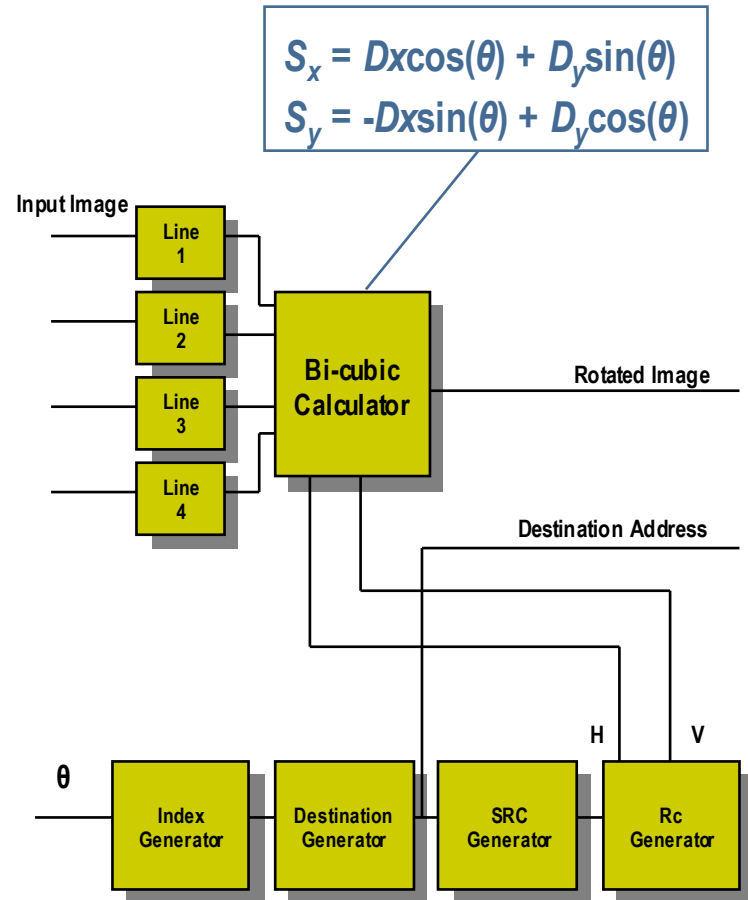
Real-Time Image Rotation

- Non-real time typically implemented using processor and frame store
- Real-time image rotation performed using bi-cubic function in FPGA
 - Pixels remapped to rotational co-ordinates



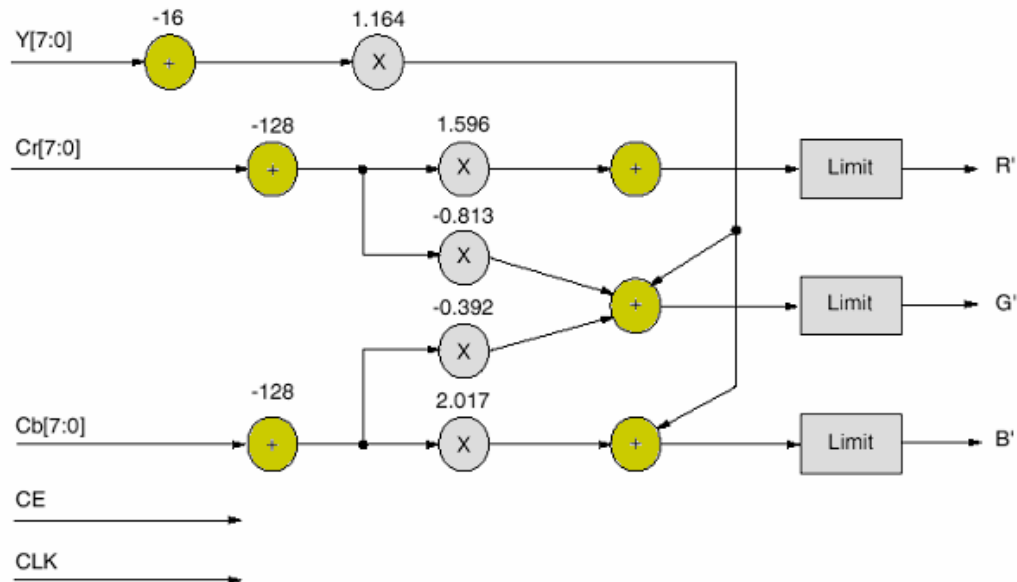
Real-Time Image Rotation

- Medical imaging example
 - 1024 x 1024 x 12 @ 30 f/s
 - 40 MHz Pixel Clock
 - 160 MHz Core Clock
- Xilinx XC2S300E FPGA
 - 12 Block RAMs for line buffers
 - 2 Block RAMs for RC lookup tables
 - 5 multiplier pixel calculation
 - Sine/cosine, 2 Block RAMs
 - Dx, Dy calculation
 - Sx, Sy calculation
 - Control



http://www.xilinx.com/products/logiccore/dsp/rotation_resize.pdf

Xilinx Color Space Converter

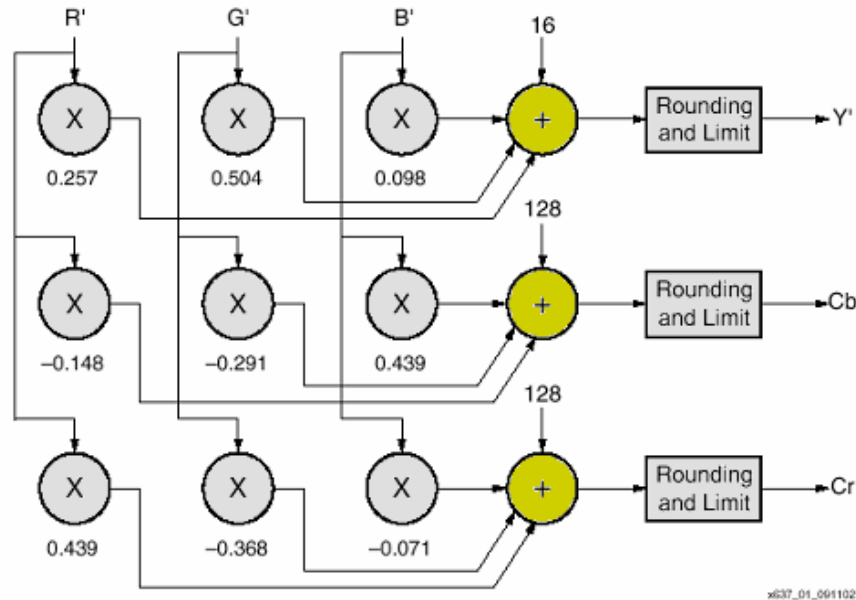


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- Free of charge Reference Design available “as is”
- For more details check out <http://www.xilinx.com/xapp/xapp283.pdf>

**Reference
DESIGN**

R'G'B' to Y'CbCr Converter



- Free of charge Reference Design available “as is”
- For more details check out <http://www.xilinx.com/xapp/xapp637.pdf>

**Reference
DESIGN**

MPEG IP for Xilinx FPGAs

IP Core or Reference Design

XAPP610 Video Compression using DCT

XAPP611 Video Compression using IDCT

XAPP208 IDCT Implementation in Virtex

2-D DCT

1-D DCT

2-D DCT

2-D DCT/IDCT

2-D IDCT

2-D DCT

2-D DCT/IDCT

DCT/IDCT

MPEG-2 SDTV I & P Encoder

MPEG-2 HDTV I & P Encoder

Reference
DESIGN

Reference
DESIGN

Reference
DESIGN

LogiCORE

LogiCORE

Alliance
CORE

Alliance
CORE

Alliance
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Alliance
CORE

Alliance
CORE

Provider

Xilinx Inc.

Xilinx Inc.

Xilinx Inc.

Xilinx Inc.

Xilinx Inc.

eInfochips

BARCO-SILEX

CAST Inc.

CAST Inc.

CAST Inc.

TILAB

Duma Video

Duma Video

XILINX

XILINX

XILINX

XILINX

XILINX

eInfochips

BARCO
SILEX

CAST

CAST

CAST

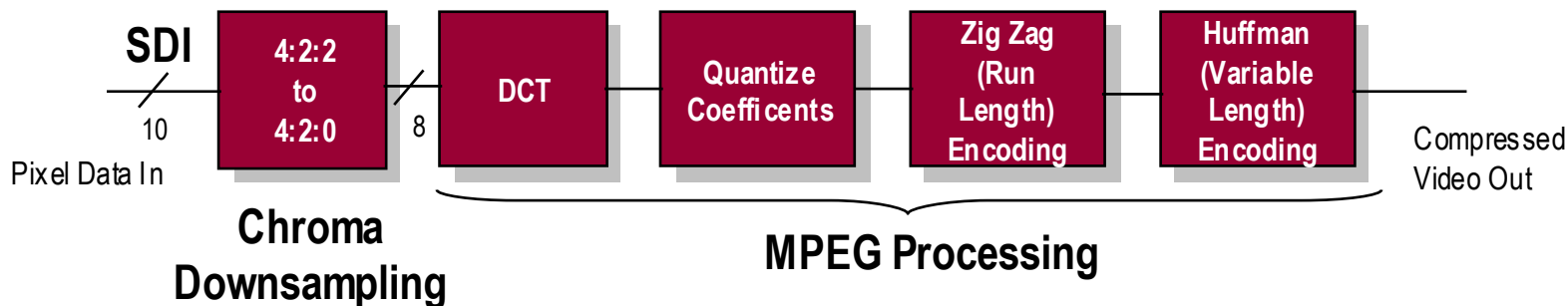
TELECOM LAB
ITALIA



See www.xilinx.com/ipcenter for more details

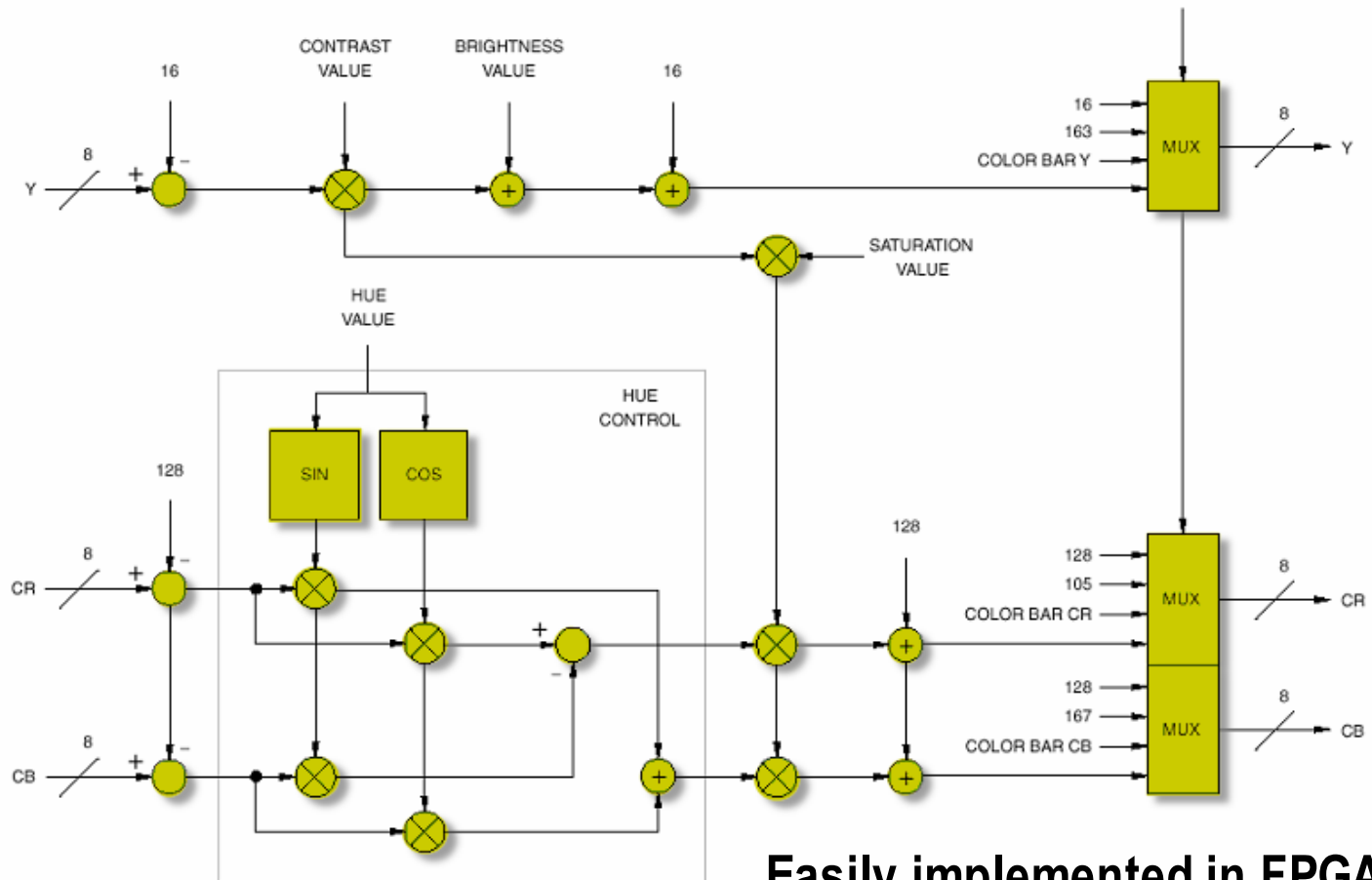
Chroma Downsampling

- Most MPEG-2 applications use 8-bit 4:2:0 sampling
- But incoming data usually 10-bit 4:2:2 video
 - Maybe via SDI (Serial Digital Interface) for example
- Conversion therefore needed before processing
 - Easy implementation using high-speed FIR filters in FPGA



Picture Enhancement

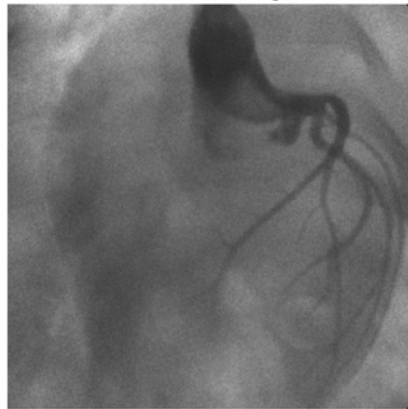
00 = BLACK SCREEN
01 = BLUE SCREEN
10 = COLOR BARS
11 = NORMAL VIDEO



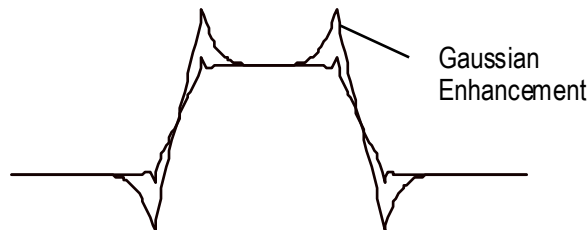
Easily implemented in FPGA

Spatial Enhancement

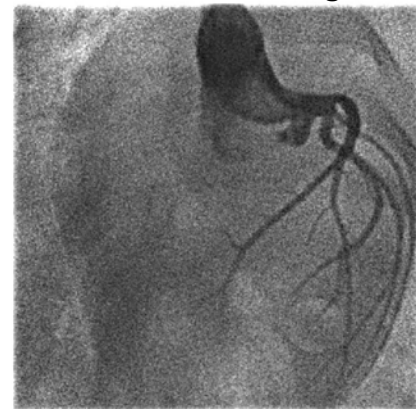
Input image



Enhanced 1D Signal

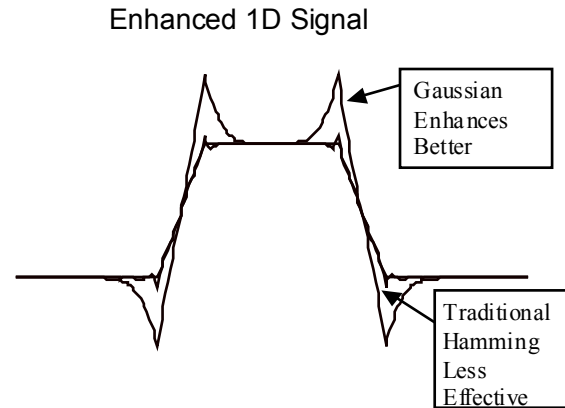
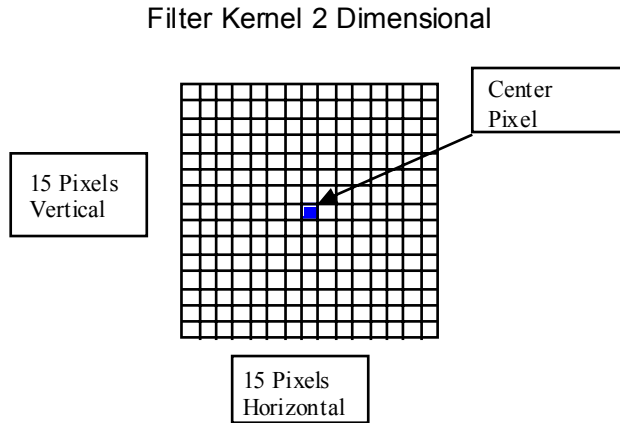


Enhanced image



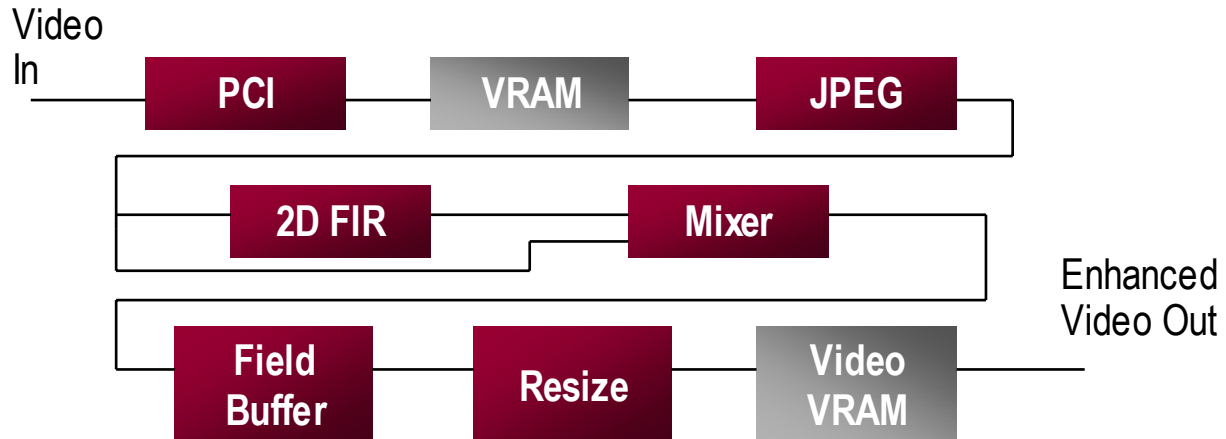
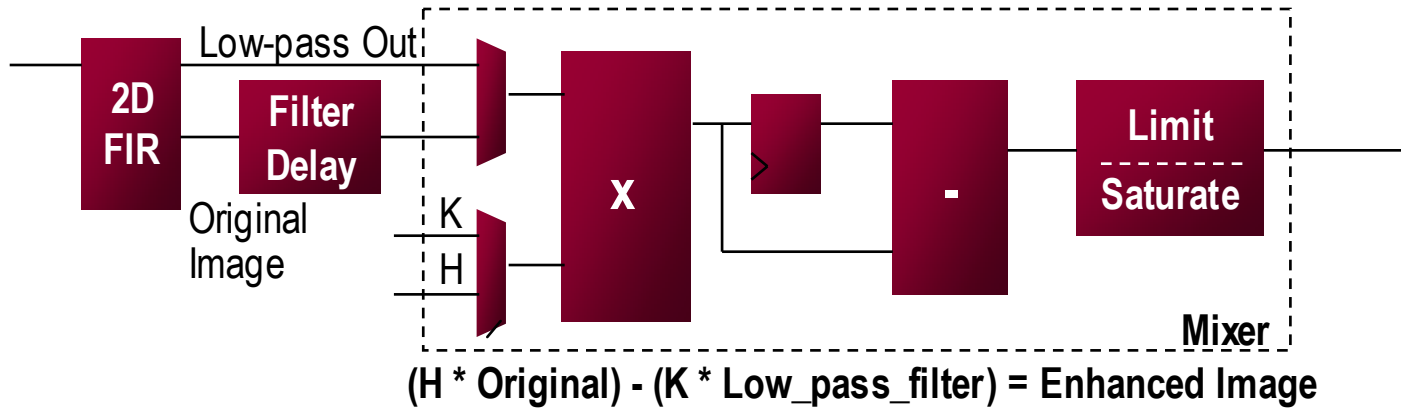
- Endless list of enhancement algorithms:
 - Spatial Filter Unsharp Masking
 - Digital Max-Detail
 - Laplacian of Gaussian Filter
 - Adaptive Histogram Equalization
 - Adaptive Kalman Temporal Filter
 - Non-Linear Median Filter
 - Non-Linear Fuzzy Filter
- Or you may have a better version of your own
 - Is there an ASSP to support your idea?
- Reprogrammable FPGAs allow for experimentation in research and development
 - Leads directly to differentiation in production

FPGAs for Spatial Enhancement



- + User selectable kernels
- + Experiment to find the filter that gives the best results (on-the-fly)
- + Parameterizable filter coefficients and windows
- + No sacrifice in performance with real-time calculations possible

Spatial Enhancement



FIR Filters for Xilinx FPGAS

- Most image filtering can be done based on two-dimensional FIR filters
 - Programmability allows experimentation with different coefficients, filter windows, etc to get the best results

IP Core or Reference Design

XAPP219 Transposed Form FIR Filters



MAC FIR



Serial Distributed Arithmetic FIR Filter



Parallel Distributed Arithmetic FIR Filter



Distributed Arithmetic FIR Filter



Provider

Xilinx Inc.



Xilinx Inc.



Xilinx Inc.



Xilinx Inc.

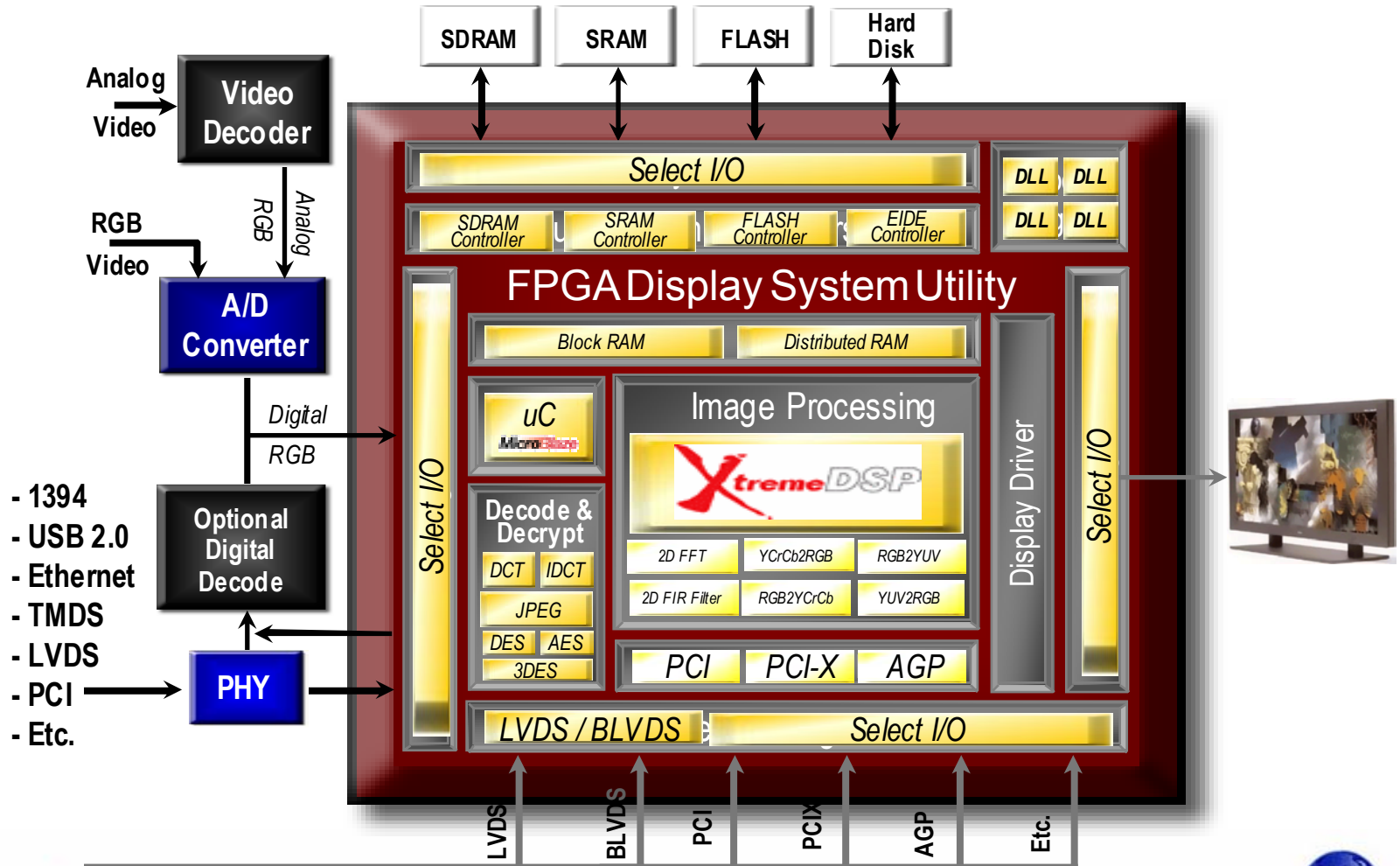


Xilinx Inc.

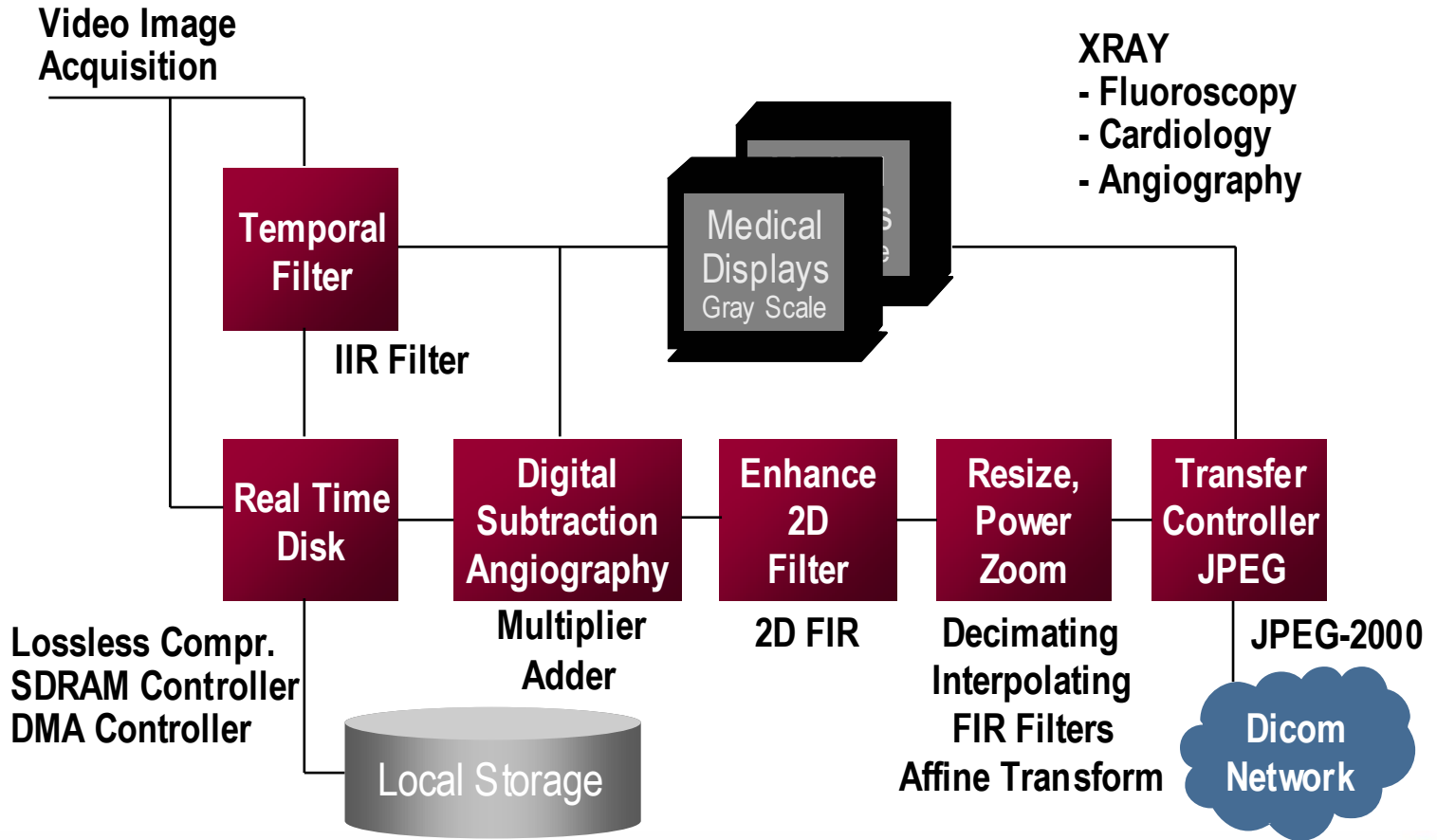


See <http://www.xilinx.com/ipcenter> for more details

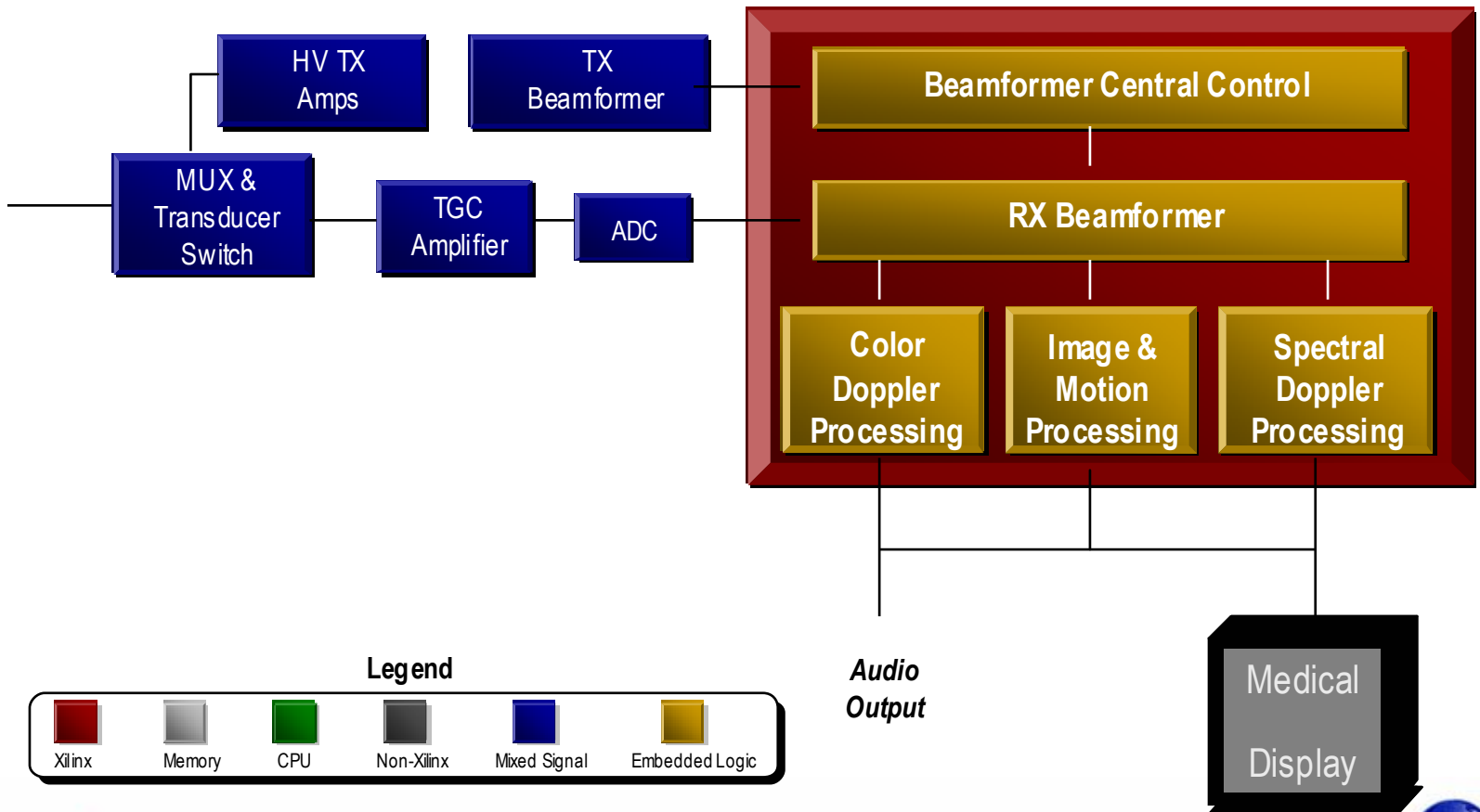
FPGA Features for Video



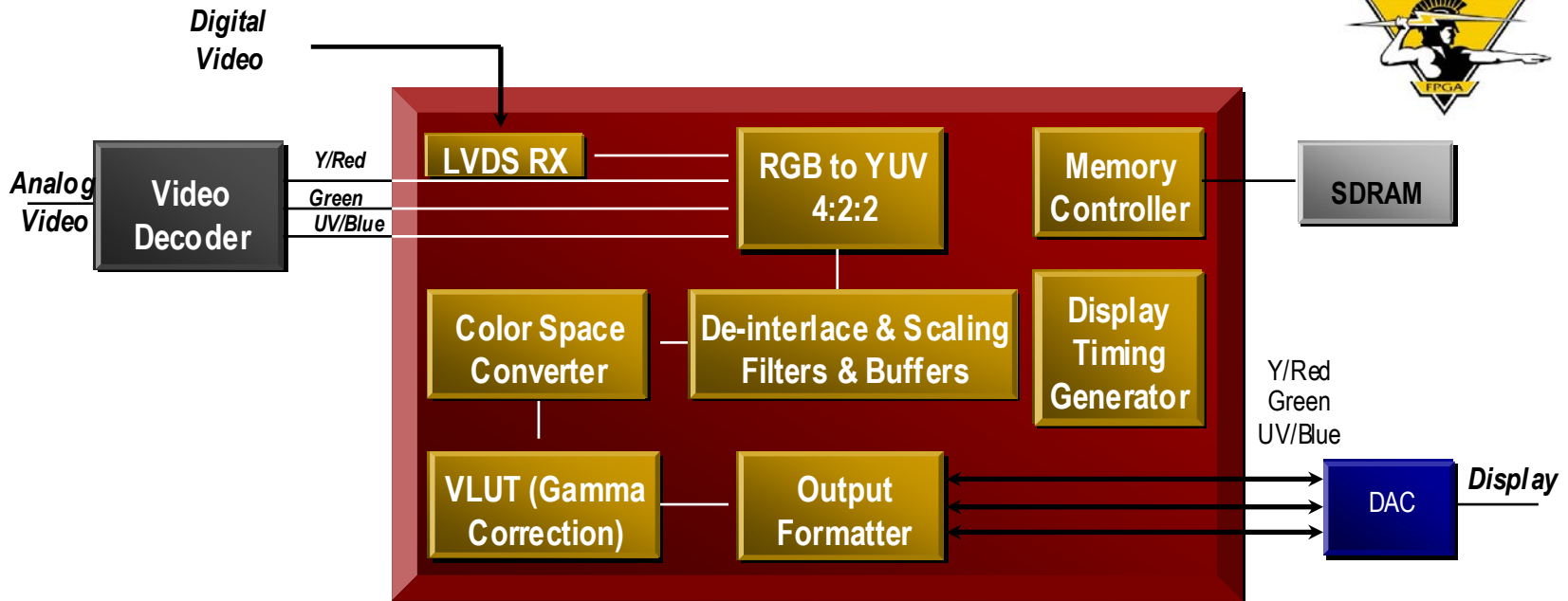
X-Ray System Example



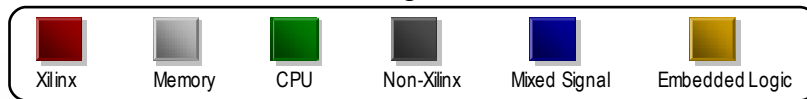
Ultrasound Beamformer



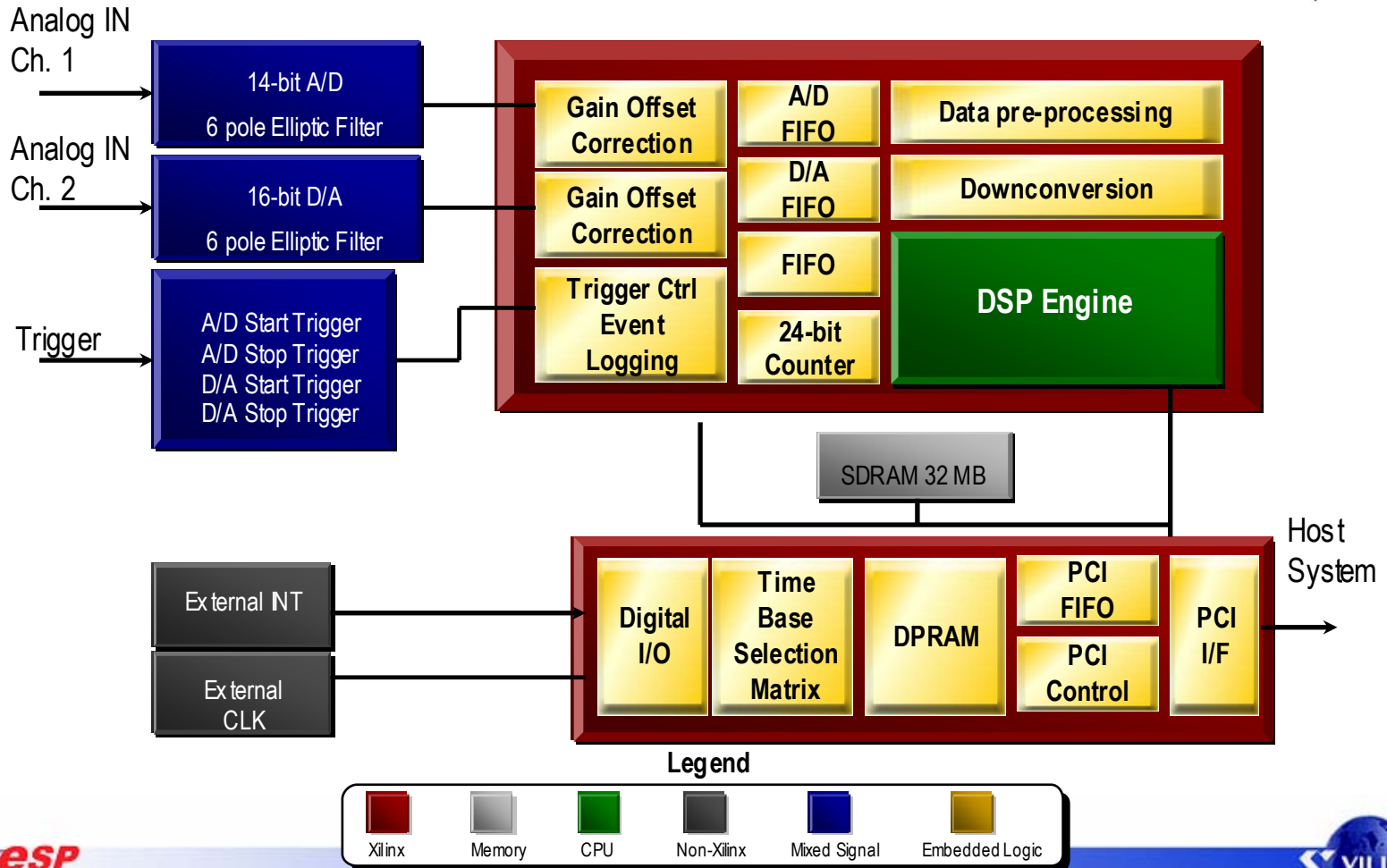
Medical Instrument Displays



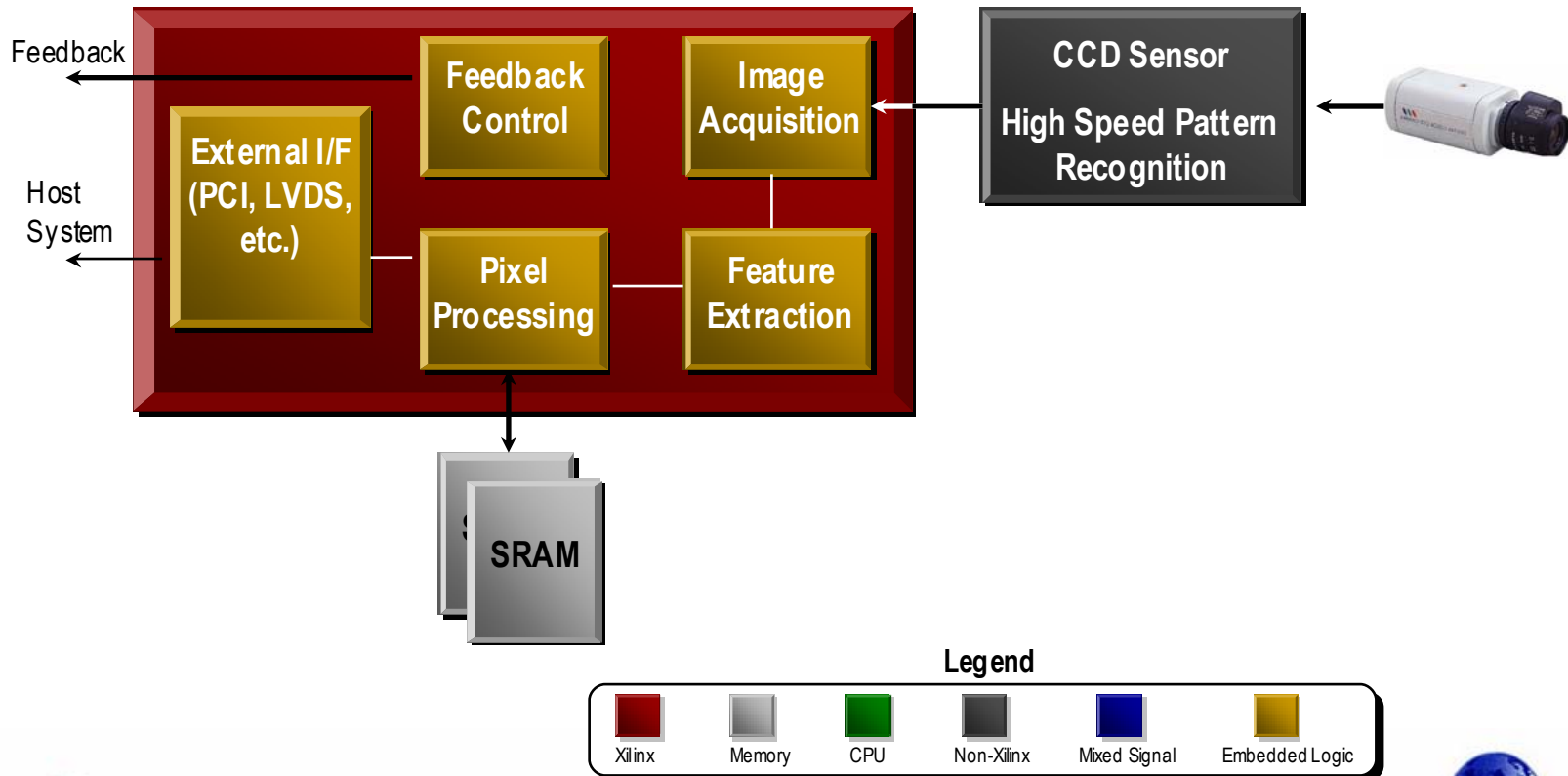
Legend



Data Acquisition Card



Smart Vision System



Digital Communications

- Remote diagnosis possible
 - Even remotely controlled procedures
 - Needs huge bandwidth for real-time analysis
- Off-site/central archiving of digital patient files
 - Requires integration of datacom/telecom interfaces
 - Storage Area Network protocols may play a role
- Xilinx supports a wide range of interfaces and protocols to enable focus on the real system value add



For more on protocols and backplanes: www.xilinx.com/esp/optical/net_tech

Xilinx - the Ultimate Connectivity Platform

LAN / MAN / WAN

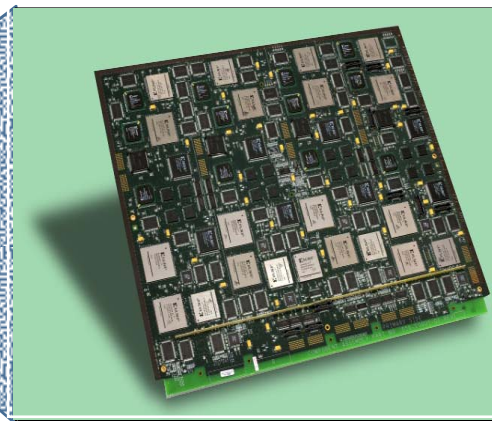
- 10/100 Ethernet
- 1Gb Ethernet
- 10Gb Ethernet
- 1Gb Ethernet PHY
- 10Gb Ethernet XAUI
- OC48* OC192* OC768*

Board-to-Board

- PCI 32/33
- PCI 64/66
- PCI-X 133
- 3GIO
- Fibre chan.
- 10GE XAUI
- RapidIO
- CSIX
- HyperTransport™
- RapidIO™ Serial
- HyperTransport™

Chip-to-Chip

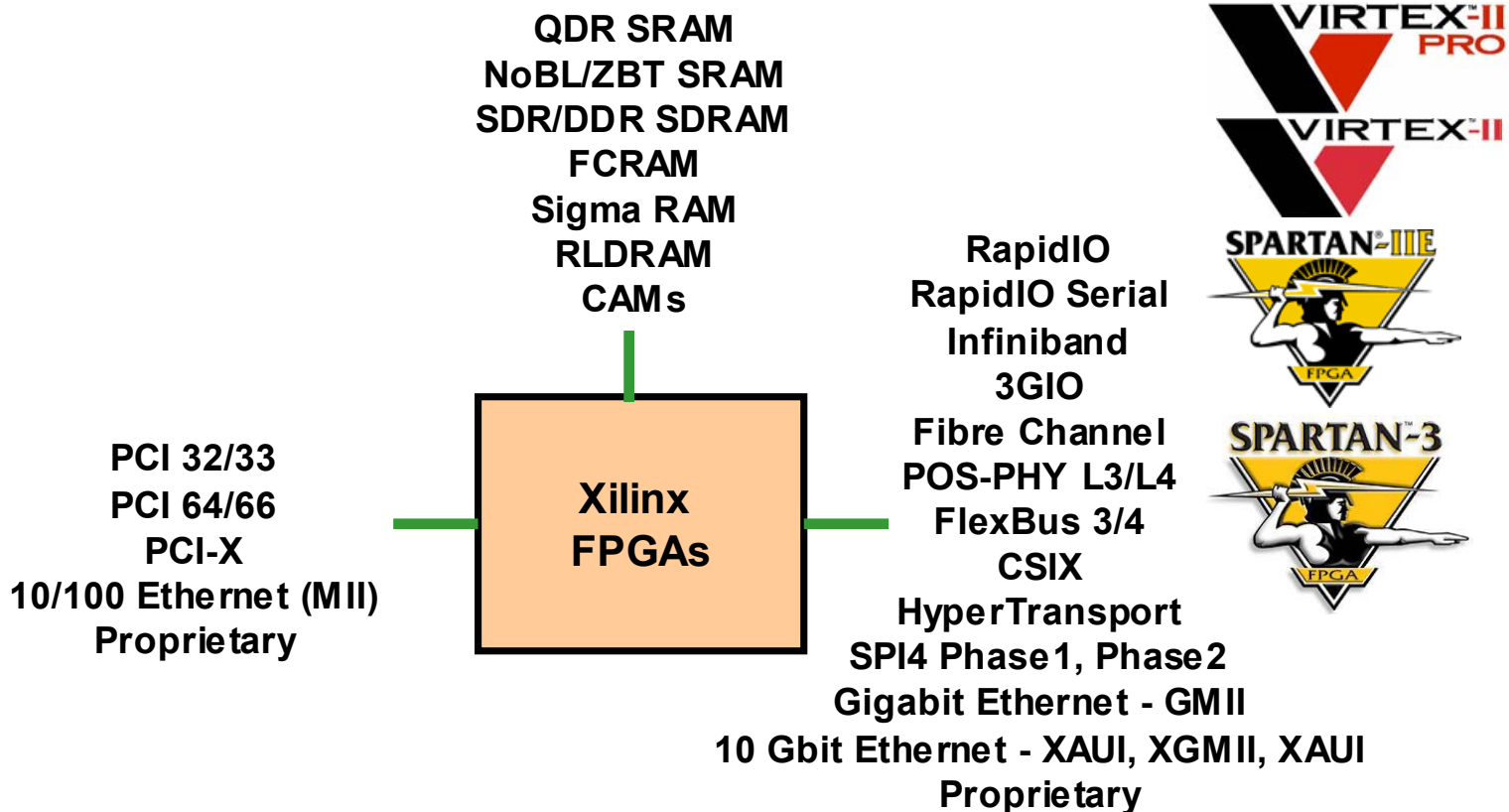
- PCI 32/33
- PCI 64/66
- PCI-X 66 & 133
- RapidIO™
- 10GE XAUI
- POS-PHY L3/L4
- Flexbus 4
- CSIX
- HyperTransport™
- 3GIO



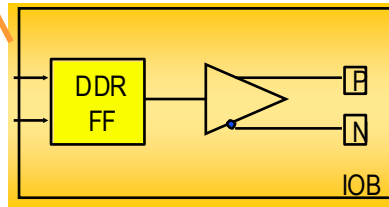
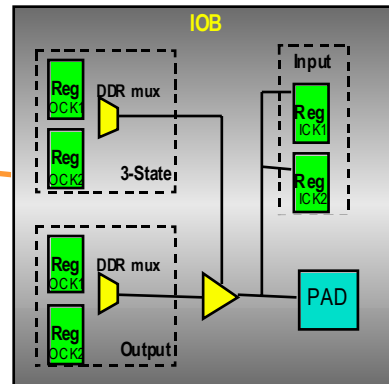
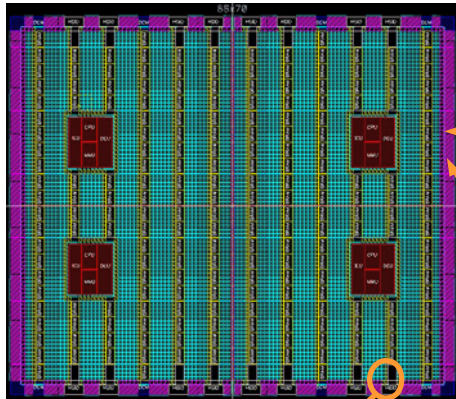
- Serial Standards – Virtex-II Pro
- Parallel Standards – Spartan-3, Spartan-IIE, Virtex-II & Virtex-II Pro

* SONET compatible, supports data rate

Supporting Legacy & Evolving System Interfaces

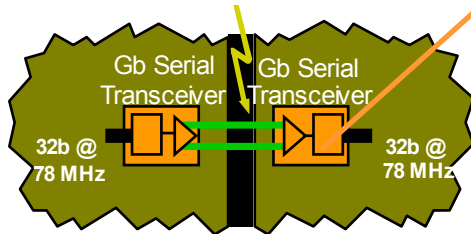


Virtex-II Pro Helps You Manage the Transition from Parallel to Serial I/Fs



Select/O-Ultra™ technology for parallel interfaces

- 25 I/O standards
- XCITE technology
- 840 Mbps LVDS
- Dedicated DDR registers



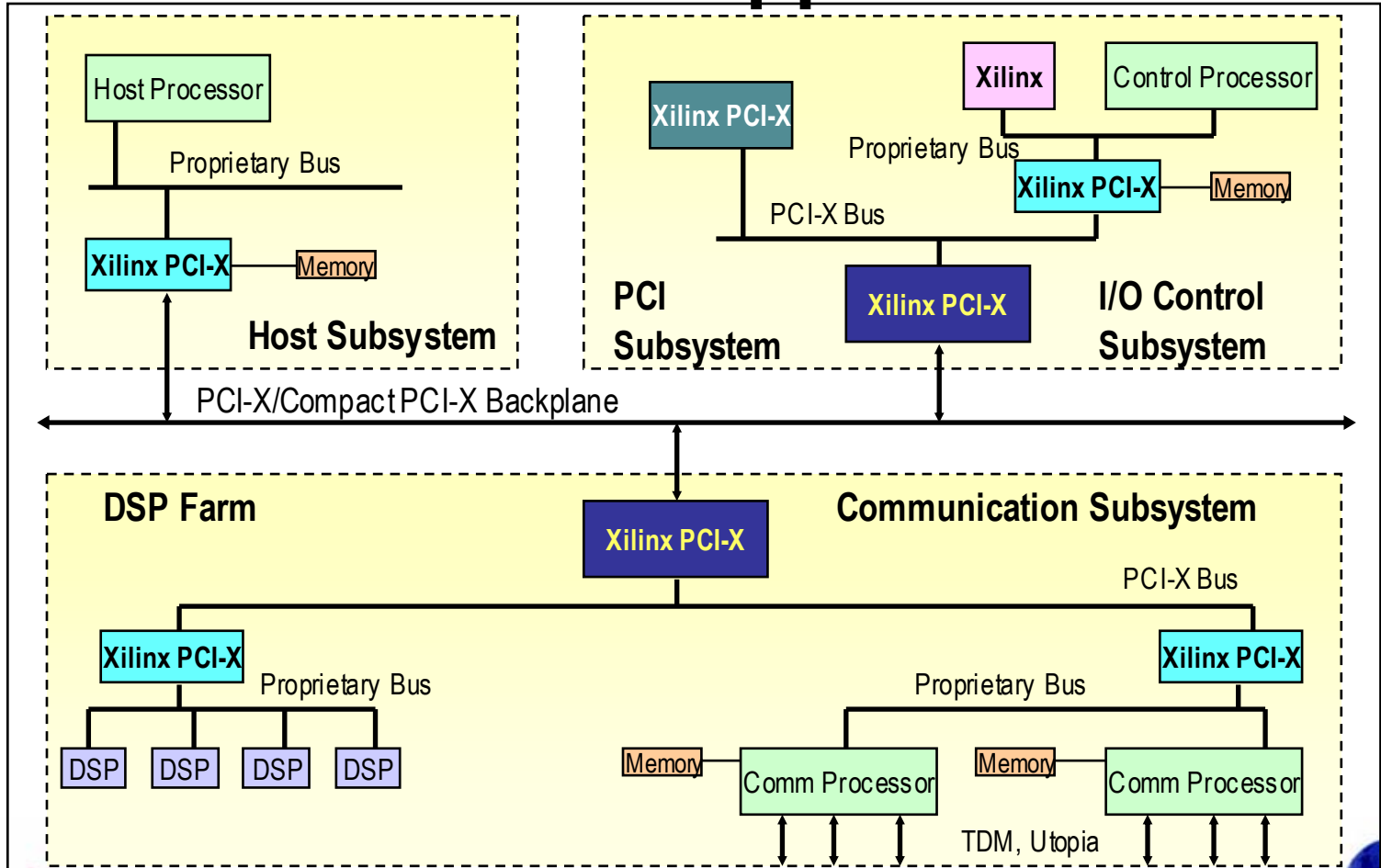
Rocket I/O Multi-Gigabit Serial Transceivers

Up to Sixteen 3.125 Gbps transceivers

- Helps preserve investment in legacy designs
- Eases transition from parallel to serial technology
- Parallel interface designs will not go away

PCI-X Markets

Embedded Applications



Lowest Cost Parallel Interconnect Solutions

- Best value in popular cores
 - PCI 32/33 effective cost below 75 cents* (\$.75)
- Physical interfaces and system elements
 - 25 I/O standards, DDR I/O registers, DCMs
- Popular IP cores
 - Pre-engineered, Drop-in functionality
 - Fully compliant, Pre-verified



The **Real-PCI™**

PCI 32/33 and 64/33

PCI 32-bit, 33 MHz under \$.75!

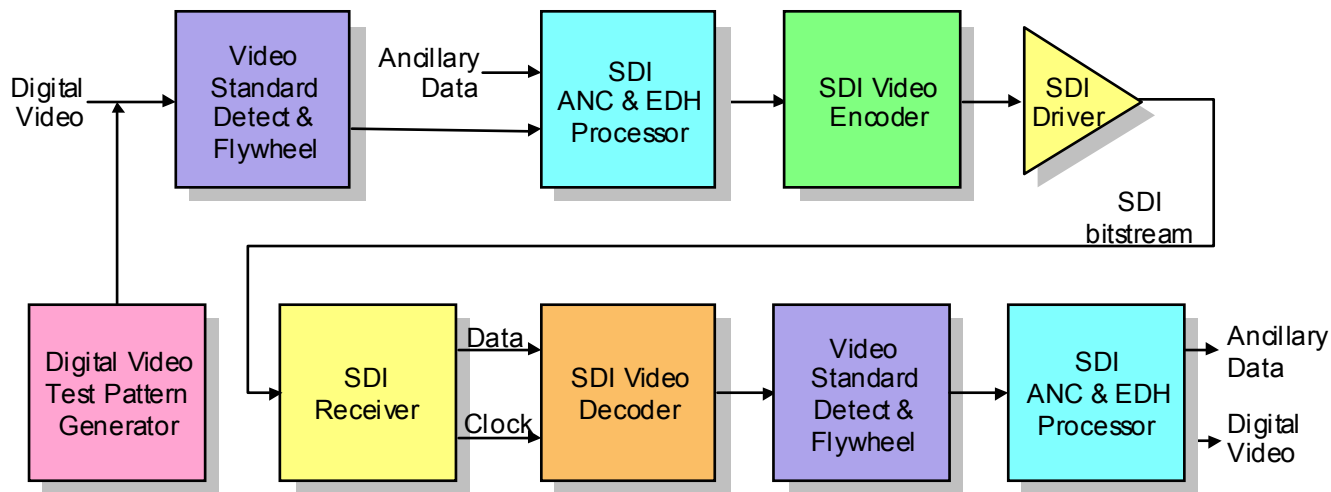


* Based on pricing for 2004, 250K units

SDI - Serial Digital Interface

- Standard for transporting *uncompressed* standard-definition digital video serially over coax cable
 - ANSI/SMPTE 259M
 - ITU-R BT.656
 - Many related SMPTE and ITU standards and recommended practices
- Xilinx has a set of application notes showing how to implement the SDI protocol
- FPGA-based solutions are *very* competitive with ASSP SDI solutions

SDI Application Notes



-  XAPP247: SDI Physical Layer Implementation
-  XAPP248: Digital Video Test Pattern Generators
-  XAPP288: SDI Video Decoder
-  XAPP298: SDI Video Encoder
-  XAPP299: SDI Ancillary Data & EDH Processors
-  XAPP625: SDI Video Standard Detector & Flywheel Decoder

Coming Soon!

<http://www.xilinx.com/xapp/xapp248.pdf>

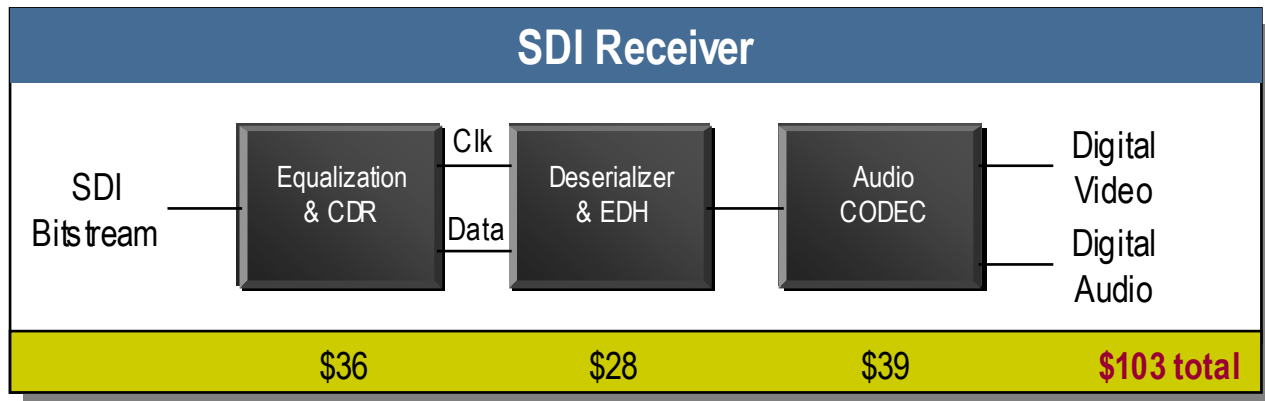
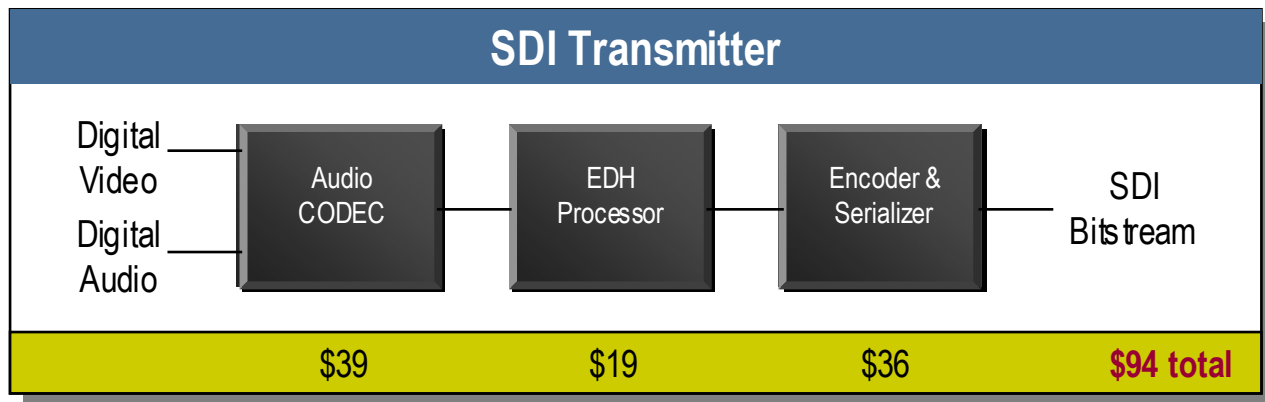
<http://www.xilinx.com/xapp/xapp288.pdf>

<http://www.xilinx.com/xapp/xapp298.pdf>

<http://www.xilinx.com/xapp/xapp299.pdf>

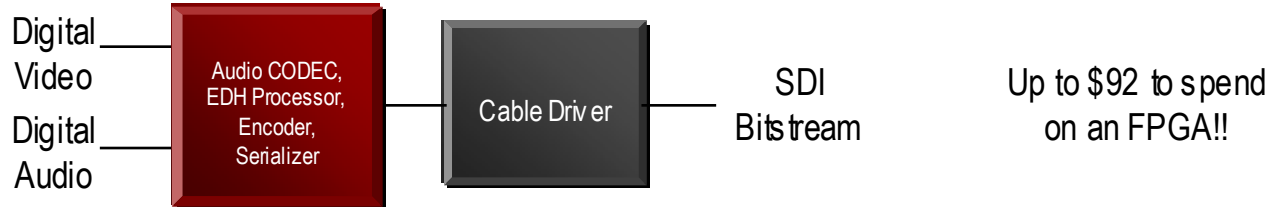
<http://www.xilinx.com/xapp/xapp625.pdf>

ASSP SDI Implementation



Xilinx SDI Implementation

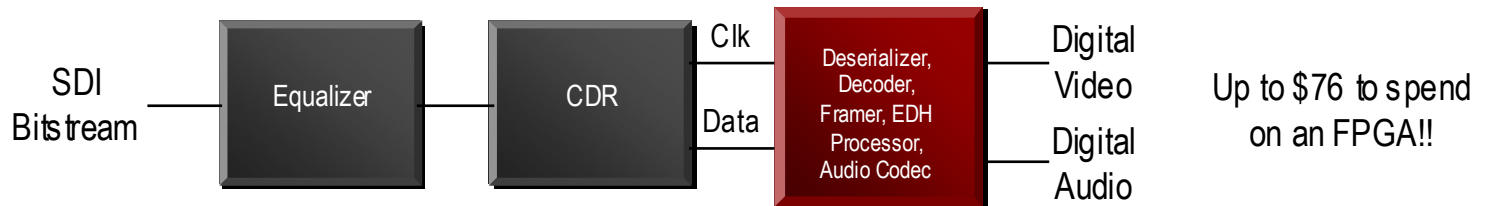
SDI Transmitter



Xilinx FPGA

\$2

SDI Receiver



\$10

\$17*

Xilinx FPGA

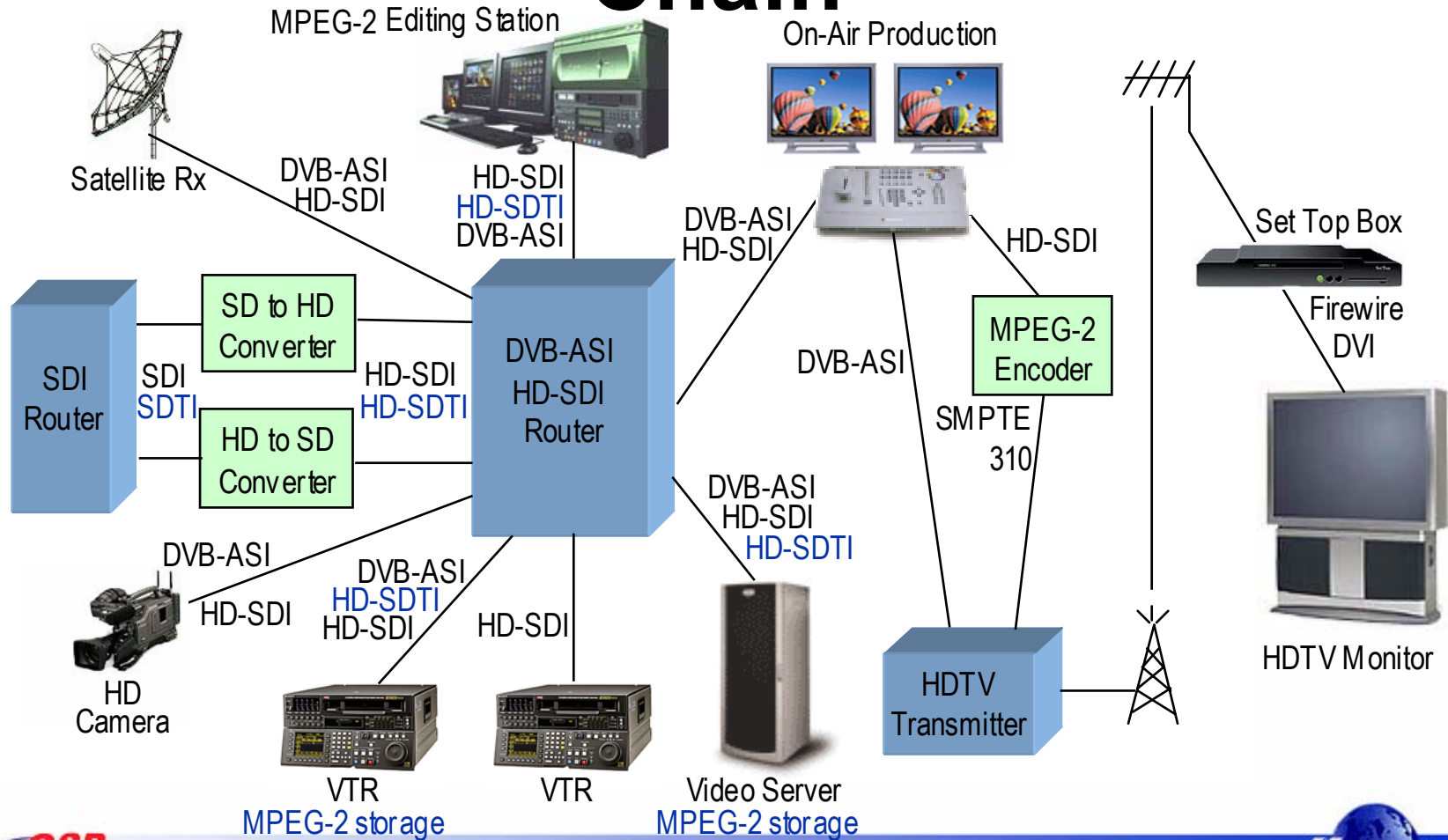
*For fixed operation at 270 Mb/s, CDR can be replaced with \$3 ASSP and a DLL in the FPGA (XAPP250)

SDI functionality integrated into existing system FPGA gives even more saving

HD-SDI - High Definition SDI

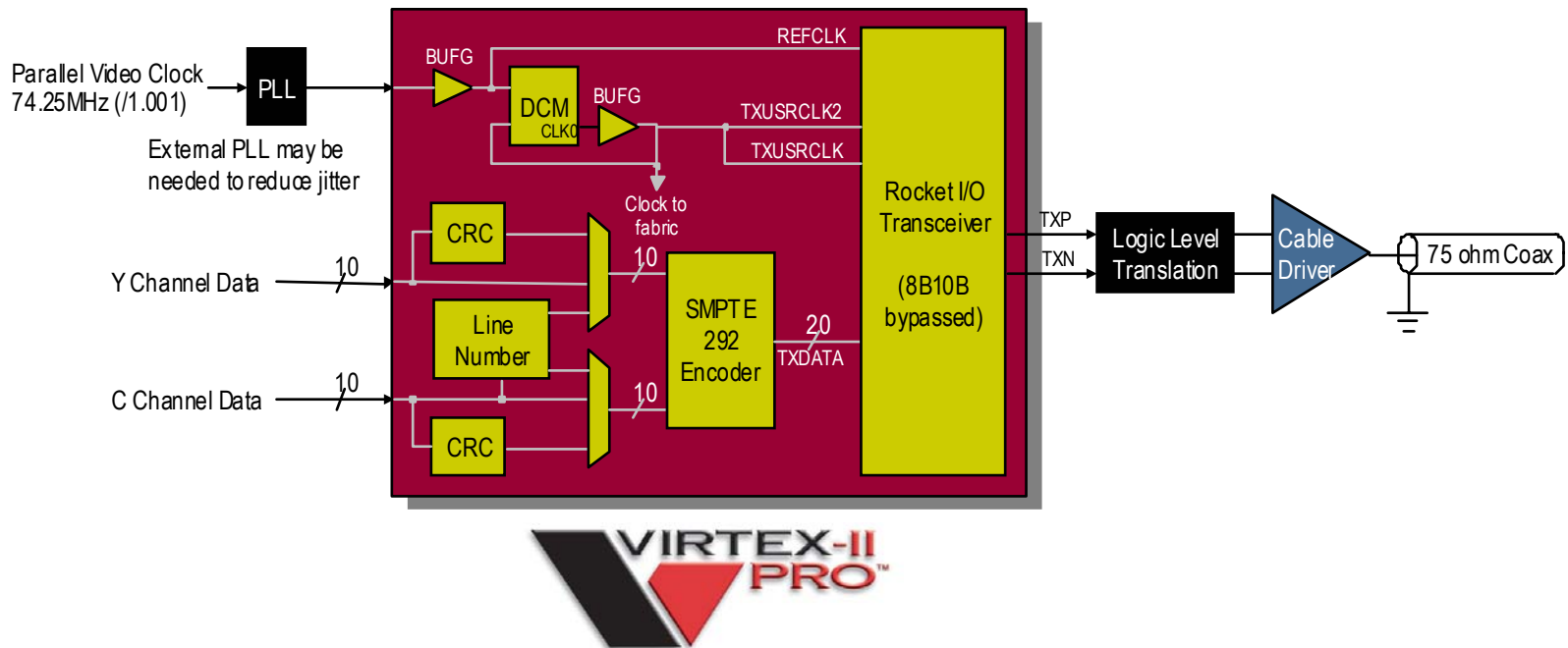
- Transports *uncompressed* high-definition video serially over video coax cable or fibre
 - 100+ meters of coax cable
 - 2km of single mode fibre
- Two bit rates
 - 1.485 Mb/s
 - 1.485/1.001 Mb/s - primarily used in North America
- Builds upon SDI experience base
 - Encoding, decoding, framing, and equalization are similar

High Def. Video Broadcast Chain



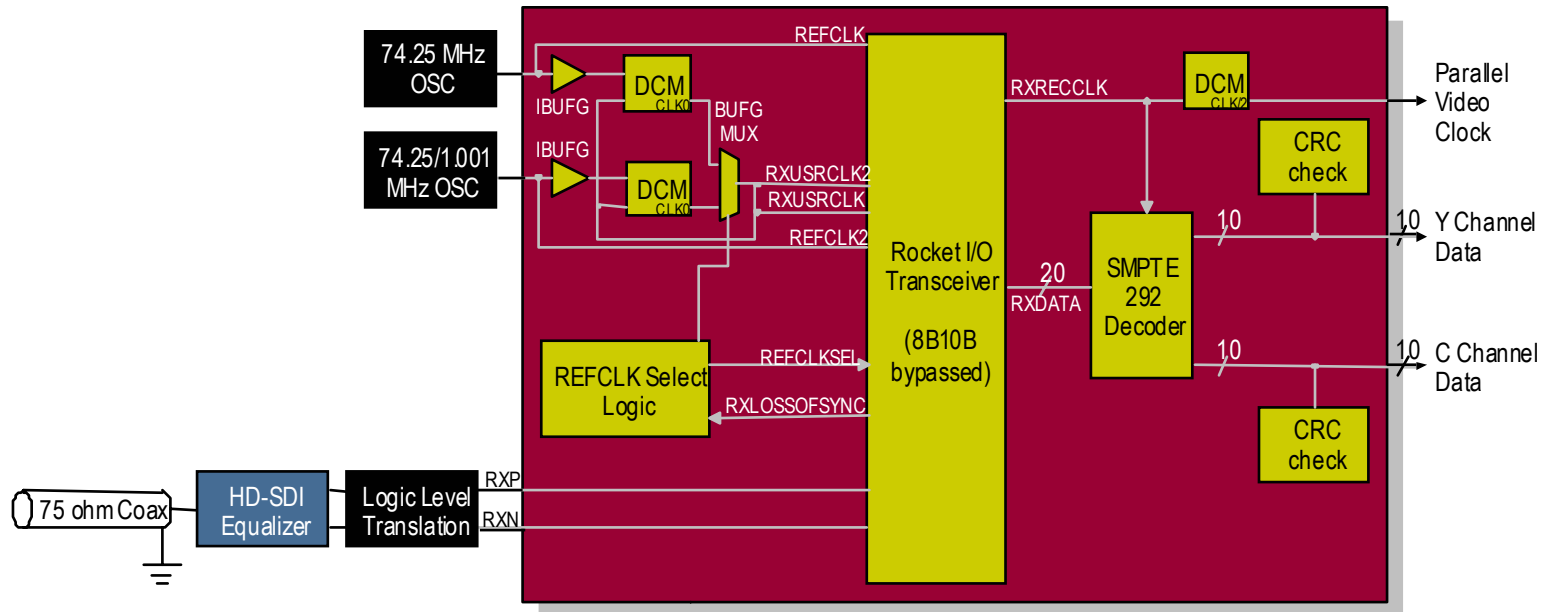
Virtex-II Pro HD-SDI Tx

Design Concept

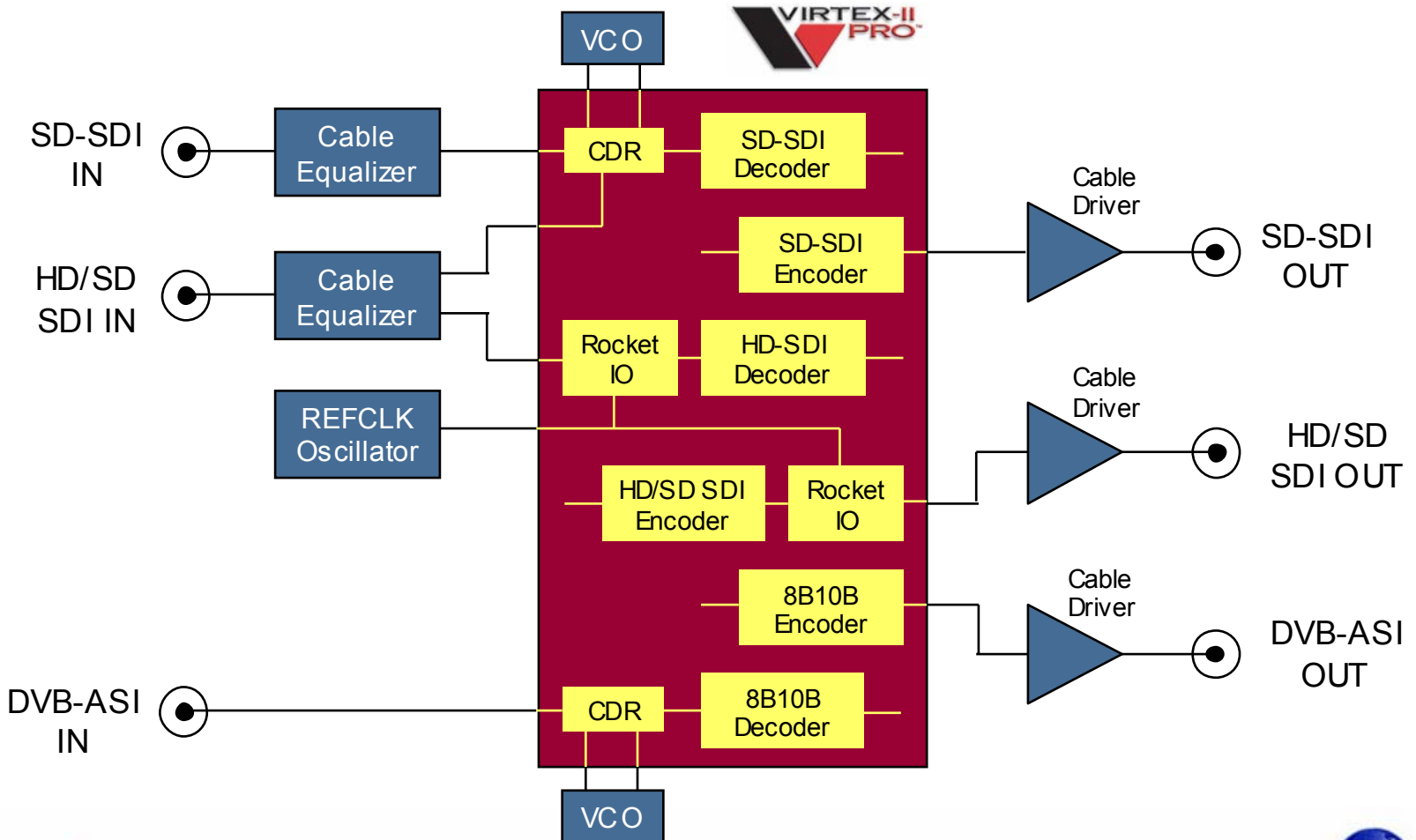


Virtex-II Pro HD-SDI Rx

Design Concept

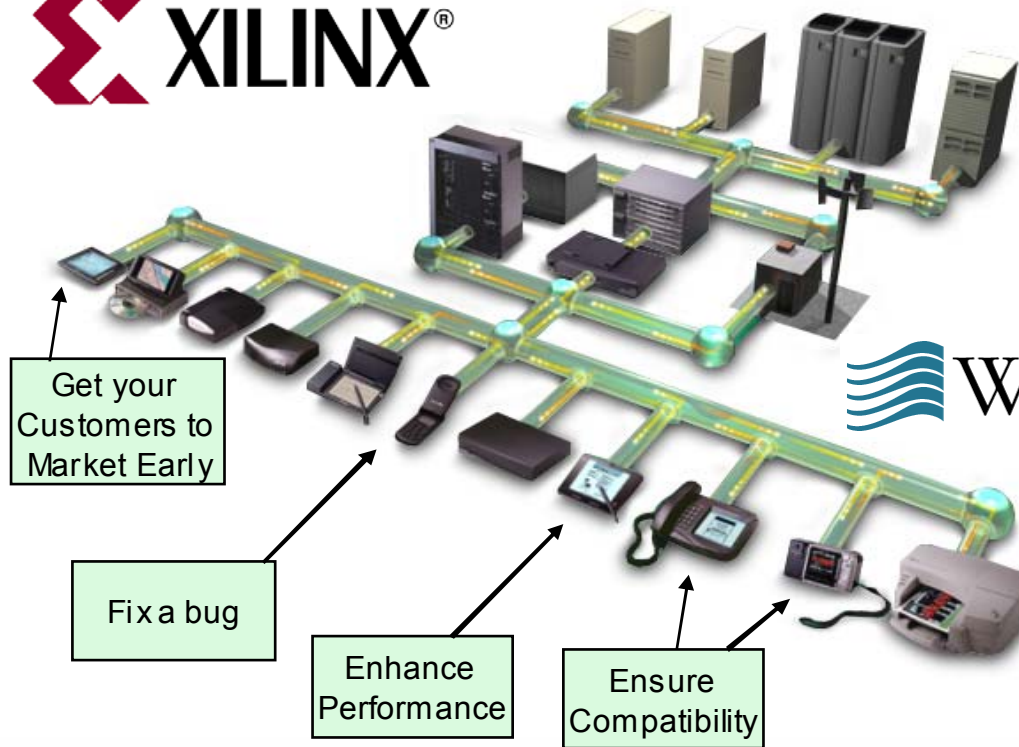


SDV Demo Board



Internet Reconfigurable Logic (IRL)

Remote update of software and hardware



Value of Remote Upgrades

- Adds key differentiation between competitive products
 - Future-proofs your products
 - Enables instant compliance to new standards
 - Continuous improvement
 - Results in increased lifetime for the gateway
 - Enable product features per end-user needs
 - Large 'bottom line' impact
 - Reduce maintenance costs
 - New Internet-based revenue opportunities
 - Increased market share
 - Get in earlier and stay in longer
 - Develop customer loyalty
- ASSPs/ASICs cannot provide this capability

Xilinx FPGAs – The Competitive Edge

- Leverage existing designs/chipset to support multiple geographies
- Faster time-to-market
- Improved inventory control
- Reduce exposure to supply issues
 - Flexibility to efficiently manage component supply problems
- React quickly to competitive pressures
 - Field upgradability
 - Bringing new features/capabilities rapidly to market

Xilinx in Scientific and Medical

- Enables the designer to add real value to the system
 - Experimentation in development leading to differentiation in production
 - Chipsets that support your *exact* requirements are never available!!
- Supports high-definition real-time processing
 - “Dedicated” logic allows for hardware acceleration of key algorithms
 - Less memory requirements for off-line processing
- System-on-chip integration saves board space and reduces BOM
 - More processing power with less devices
 - Reduction of DSP “farms” using parallel processing capability of FPGA
 - Integration of network and backplane interfaces, display controllers, level translators, clock management, etc.

More white papers on Xilinx FPGAs for image filtering, enhancement and scaling/rotation at
http://www.xilinx.com/xlnx/xil_prodcat_product.jsp?title=literature_training#conference



Questions?

espteam@xilinx.com

